



**CL-PID :OP-AMP APPLICATION TRAINER
KIT
EXPERIMENTAL MANUAL**

... A MESSAGE FROM

Today's system designers are faced with tomorrow's problems. CONTROL & INSTRUMENTATION is one of the important subject need to teach while learning electronics.

It is our vision to provide you with the product you need for training ensuring lasting reliability & quality.

OUR MOTTO;

-Light years ahead – refers to leadership.

As leaders in our industry in India, we are totally committed to servicing as the standard against which all are measured in the areas of

• Design • Quality • Value • Delivery • Support.

We are truly light years ahead of our competition in this area. That means that you our valued customers are guaranteed satisfaction.

As you will move through this manual you will quickly discover that we have complete, truly innovative & superior training products. We are so committed to quality that we back our products with a complete comprehensive warranty.

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SAFETY RULES

Carefully follow the instructions contained in this manual as they provide you with important points on safety during the installation, use and maintenance. Keep this manual always with you for easy reference.

Arrange all accessories in order after unpacking, so that their integrity is checked with respect to its checklist. Also ensure that no visible damage as such appear on any accessories.

In CL-PIDkit, signals are to be monitored with an oscilloscope as explained in the manual. For observation of signals on oscilloscope, use 1:10 Attenuation Probe.

These kits must be employed only for the use for which it has been conceived, i.e. as educational equipment, and must be used under the direct survey of expert personnel. Any other use is improper and so dangerous. The manufacturer cannot be considered responsible for eventual damages due to improper, wrong or unreasonable uses.

In case of any fault or malfunctioning in the trainer kit, turn off the power supply and do not tamper the kit. In case servicing is required, contact the service center for technical assistance.

The kits are liable to malfunction/under-perform if they are not operated under standard environmental conditions of temperature and humidity.

WARRANTY

These kits are warranted against defects in workmanship and materials. Any failure due to defect in either workmanship or material should occur under normal use within a year from the original date of purchase, such failure will be corrected free of charge to the purchaser by repair or replacement of defective part or parts. When the failure is result of user's neglect, natural disaster or accident, we charge for repairs regardless of the warranty period. The warranty does not cover perishable items like connecting chords, Crystals, etc. and other imported items. No warranty for mechanical damages after delivery.

This warranty is subject to the following conditions and limitations. The warranty is void and inapplicable if the defective product is not brought or sent to our authorized service center or sales outlet within the warranty period. Defective product is, on Akademika Lab Solutions sole judgment. The defective product will be replaced with new one or repaired without charge or with charge.

In the warranty period if the service is needed, purchaser should get in touch with the service center or sales outlet. The purchaser should return the product to the service center or sales outlet at his or her sole expense. The loss and damage in transit will be outside the preview of this warranty. A returned product must be accompanied by a written description of the defects. Type and Model No. of kits has to be mentioned specifically. We return the product to the purchaser at our expense. In case that warranty does not cover the product on Akademika Lab Solutions judgment, we repair the product after obtaining prior permission from purchaser who receives an estimate statement about repairing charges. In such case, Akademika Lab Solutions bares the transporting expenses required to send back all the repaired products for the moment, and then repairs and transporting expenses will be charged against the purchaser by the statement of accounts.

When the authorized sales agents sell our products, they must notify the purchaser of the warranty contents, but have no rights to stretch the meaning of original warranty contents or offer additional warranty. Akademika Lab Solutions does not provide any other promise or suggestive warranty and hold no liability for the damage caused by negligence, abnormal use or natural disaster. Akademika Lab Solutions is not responsible for the damages even if it is notified of above dangers in advance as well.

For more special service or overall repairs, maintenance and up-gradation of products, be sure to contact our service center or sales outlet.

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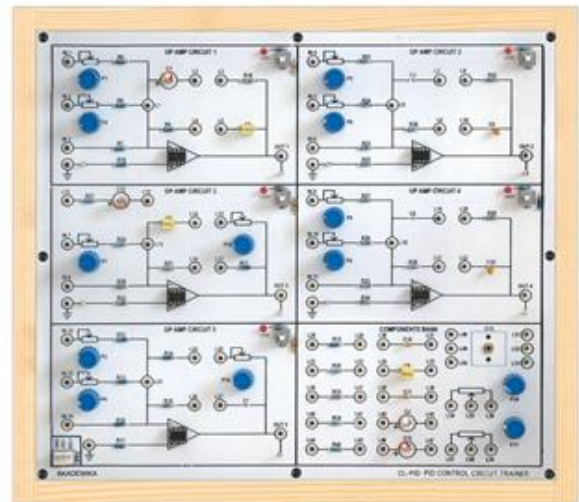
TECHNICAL SPECIFICATIONS

- In Built power supply.
- +5V/3A, +12V/1A, -12V/0.5A, -5V/0.5A, -12V TO +12V/0.5A.
- On-bard Capacitor and Resistor bank.
- Different Nonlinear Amplifiers.
- Group of 8 Operational Amplifiers in Op-Amp unit.

- **Deliverables** :

1	Power Supply	: 01no.
2	Power Chord	: 12 no.
3	Short Links	: 12 no.
4	Long Links	: 01 no.
5	Experiment Manual	: 01 no.
6	3 pin Female to Female CPU Lockable Cable	: 01 no.

FUNCTIONAL BLOCK



INTRODUCTION

“Automatic Control Principle” is an important professional course in automatic control, industrial automation, electric automation, instrument and measurement, machinery, dynamics, metallurgy, etc. It is also the main course of corresponding specialties in our domestic academies.

Currently, the trend of technology development is highly integration also highly differentiation, it requires our college students not only to be of solid theoretical foundation, but also have strictly engineering technology training, incessant improving experiment research ability, analyzing and calculating ability, summarizing and concluding ability as well as the ability to solve all kinds of practical problems. 21 century requires us to cultivate “creation type, exploitation type, application type” talents; it presents a new test of our experiment teaching. The theoretical property of Automatic Control Principle course is strong, so when we learn this course, setting necessary experiments will be benefit for students to deepen understanding, further master basic theory and analyzing method, cultivate students’ ability of analyzing and solving problems, and making the nonfigurative conception & theory imaginable and specific, enhancing the students’ learning interests. Finishing the experiment of the course is an important teaching assistant link.

CL-PID:Op-Amp Application Trainer KIT of our company is special developed for the course Automatic Control Principle; it is a useful tool for teacher and student’s scientific research. It has the advantages that visual and flexible operation so that it is easy to cultivate students’ experiment skill, in order to fully develop students’ independent thinking ability and subjective initiative, the book requires definitely that doing related theory calculation and analysis before experiment, and the steps of experiment are always principled. The main problems we may facing during experiment are listed in Thinking Question in order to attract students’ attention.

The apparatus can finish ten experiments such as linear continuous system, nonlinear system, sampling system and status feedback (pole assignment), etc. After the experiment, students must finish detailed experiment report, including experiment methods, experiment process, results and experience, etc.

As the level of compiler is limited, so it is hard to avoid mistakes and misgivings in the book, we sincerely hope to get the criticism and correction from readers.

SUMMARY OF EXPERIMENT SYSTEM

Experiment System Composition

CL-PID experiment apparatus is mainly composed by power supply unit, signal unit, chip unit, nonlinear unit, operational amplifier unit, and signal sampling and restoration unit.

Power supply unit includes power supply switch, +5V, -5V, +12V, -12V and -12V—+12V adjustable voltage output.

Chip unit provides capacitances used in experiment, resistance and adjustable resistance, convenience for students' experiment.

Nonlinear unit includes three groups of different typical nonlinear unit.

Operational amplifier unit includes eight groups of different setting parameters' operational amplifier units.

The Whole Layout and Part Module Frame of Experiment Kit

(1) The PCB1 layout of the experiment kit is showed as the following figure 1:

NONLINEAR UNIT 1	NONLINEAR UNIT 2	POWER SUPPLY UNIT
NONLINEAR UNIT 3	OP AMP CIRCUIT 6	
OP AMP CIRCUIT 7	OP AMP CIRCUIT 8	

Figure 1: the PCB1 layout of CL-PID: OP-Amp Application Trainer Kit.

(2) The PCB2 layout of the experiment kit is showed as the following figure 1:

OP AMP CIRCUIT 1	OP AMP CIRCUIT 2
OP AMP CIRCUIT 3	OP AMP CIRCUIT 4
OP AMP CIRCUIT 5	COMPONENTS BANK

Figure 2: The PCB2 layout of CL-PID: Op-Amp Application Trainer Kit.

(2) Operational amplifier unit uses LM741 operational amplifier, circumscribed different resistance, capacitance, variable potentiometers to form a circuit. Operational amplifier zero setting adjustable resistance has been linked into circuit and finished zero setting. Using leads to circumscribe different resistance and capacitance can form different circuit to finish experiment.

(3) Nonlinear unit use LM741 operational amplifier, circumscribed different resistance, capacitance, variable potentiometers and two-way voltage regulator tube to form a circuit. Operational amplifier zero setting adjustable resistance has been linked into circuit and finished zero setting, resistance, capacitance and two-way voltage regulator tube and other components have been fixed, use leads to link input signal to finish experiment.

(4) Chip unit is formed of resistance, capacitance, variable potentiometer with different parameters, leads are all well linked, only need to circumscribe lead to finish the experiment.

Resistors Bank

PCB1

NONLINEAR UNIT 1 1. R13-10K 2. R14-10K 3. R15-10K 4. R16-10K	NONLINEAR UNIT 2 1. R17-10K 2. R18-10K 3. R19-10K 4. R20-10K	POWER SUPPLY UNIT
NONLINEAR UNIT 3 1. R21-10K 2. R22-10K 3. R23-10K 4. R24-10K 5. R25-10K 6. R26-10K 7. R27-10K 8. R28-10K 9. R29-10K	OP AMP CIRCUIT 6 1. R9-200K 2. R39-100K 3. R40-100K 4. R41-100K 5. R42-100K 6. R46-100K 7. R47-100K	
OP AMP CIRCUIT7 1. R11-200Ω 2. R30-10K 3. R31-10K 4. R32-10K 5. R33-10K 6. R34-20K	OP AMP CIRCUIT8 1. R8-100K 2. R10-200K 3. R12-510K 4. R35-100K 5. R36-100K 6. R37-100K 7. R38-100K	

PCB2

OP AMP CIRCUIT 1 1. R6-10K 2. R7-10K 3. R8-10K 4. R9-10K 5. R10-10K 6. R18-51K	OP AMP CIRCUIT 2 1. R21-100K 2. R22-100K 3. R23-100K 4. R24-100K 5. R25-100K 6. R36-200K
OP AMP CIRCUIT 3 1. R11-10K 2. R26-100K 3. R30-100K 4. R31-100K 5. R32-100K 6. R37-1K	OP AMP CIRCUIT 4 1. R27-100K 2. R28-100K 3. R29-100K 4. R33-100K 5. R34-100K 6. R38-0.5MΩ
OP AMP CIRCUIT 5 1. R12-10K 2. R13-10K 3. R14-10K 4. R16-10K 5. R17-10K 6. R19-51K	COMPONENTS BANK 1. R15-10K 2. R20-51K 3. R35-100K 4. R39-0.5MΩ 5. R40-1MΩ

EXPERIMENT

NO.1

EXPERIMENT NO:1

NAME:-

Typical Link Circuit Simulation.

OBJECTIVE:-

- Study the simulation circuit which is composed by typical linear parts.
- Research on the influence of resistance & capacitance's parameters on typical linearlink step response.
- Learn the measurement method of typical linear link step response, and learn how to calculate typical linear link's transfer function according to step response curve.

APPARATUS:-

1. CL-PID:Op-Amp Application Trainer Kit.
2. Double-trace low frequency slow scanning oscilloscope.
3. Digital Multi-meter.
4. Connecting Links.

THEORY:-

Experimental Contents

- Finish proportional component circuit simulation experiment, and research parameter variation's influence on step characteristics.
- Finish integral link simulation experiment, and research parameter variation's influence on step characteristics.
- Finish proportional integral link circuit simulation experiment, and research parameter variation's influence on step characteristics.
- Finish proportional differential link circuit simulation experiment, and research parameter variation's influence on step characteristics.
- Finish inertia link circuit simulation experiment, and research parameter variation's influence on step characteristics.
- Finish proportional integral and differential loop circuit simulation experiment, and research parameter variation's influence on step characteristics.

Experimental Principles

- **Proportional Link:-**

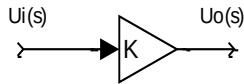


Figure 1.1.1

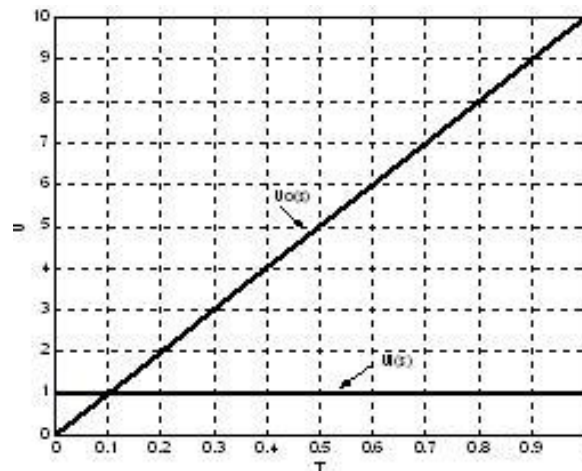


Figure 1.1.2

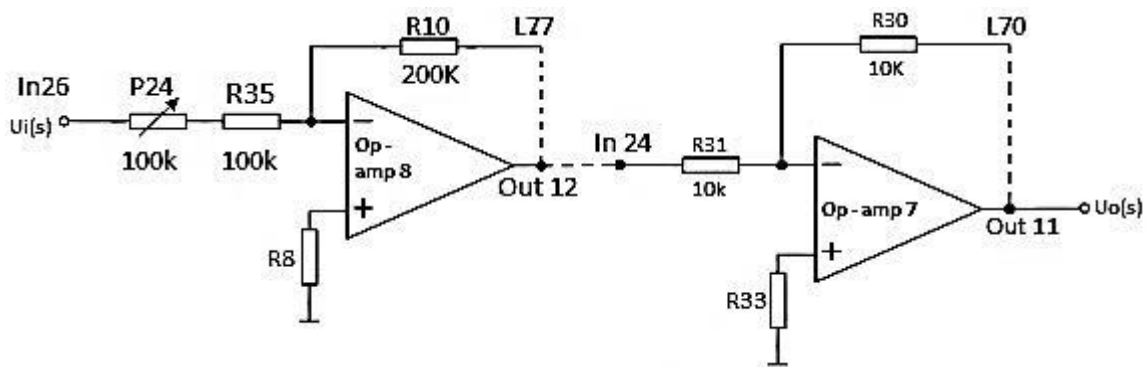


Figure 1.1.3

Proportional component's transfer function: $\frac{U_o(s)}{U_i(s)} = K$.

Proportional component's diamonds chart, step response chart and simulation circuit chart are showed as Figure 1.1.1, Figure 1.1.2, and Figure 1.1.3. $K = \frac{R_2}{R_1 + R_0}$, parameters of experiment $R_2=200K$, $R_1=100K$, $R_0=100K$, $R=10K$ or $100K$.

$$G(s) = K$$

$$K = \frac{R_2}{R_1 + R_0} = \frac{200K}{100K + 100K} = 1$$

$$G(s) = \frac{C(s)}{R(s)} \dots\dots\dots [\because H(s) = 0]$$

But i/p Signal is unit step signal.

$$\therefore R(s) = \frac{1}{s}$$

$$\therefore G(s) = \frac{c(s)}{R(s)} = 1$$

$$\therefore c(s) = R(s)$$

$$\therefore c(s) = \frac{1}{s}$$

Taking Laplace Inverse Transform of both sides

$$C(t) = u(t)$$

- **Integral Link:-**

Integral link's transfer function: $\frac{U_o(s)}{U_i(s)} = \frac{1}{TS}$.

Integral link's diamonds chart, step response chart and simulation circuit chart are showed as following Figure 1.2.1, Figure 1.2.2 and Figure 1.2.3.

$T = (R_0 + R_1)C_1$, parameter in the experiment $R_0=100K$ adjustable, $R_1=100K$, $C_1=1\mu F$, $R = 10K$ or $100K$.

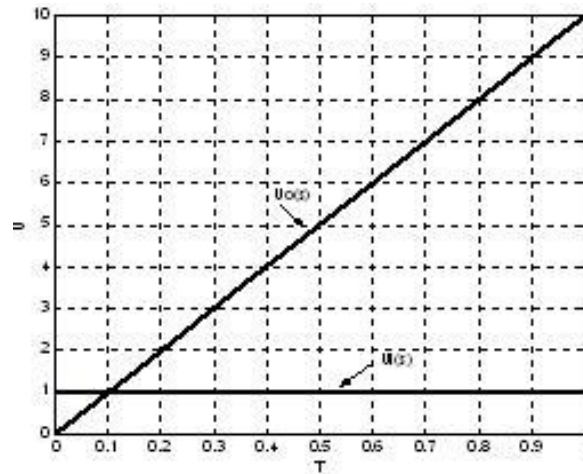
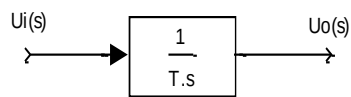


Figure 1.2.1

Figure 1.2.2

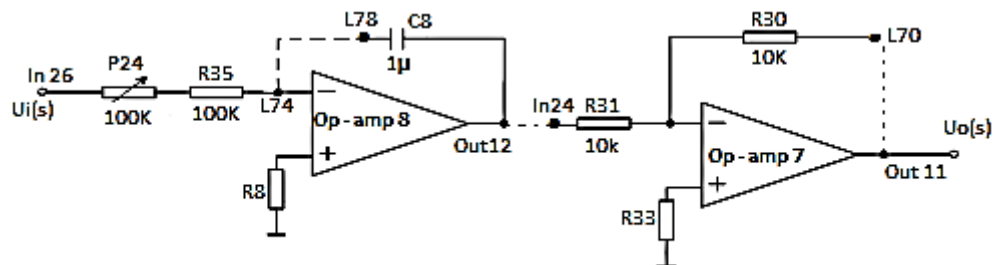


Figure 1.2.3

$$G(s) = \frac{1}{T.s}$$

$$T = (R_0 + R_1)C_1 = 200K \times 10^{-6} = 0.2$$

$$\therefore G(s) = \frac{c(s)}{R(s)} = \frac{1}{0.2s}$$

$$\therefore c(s) = \frac{5}{s} \cdot R(s)$$

$$\therefore c(s) = \frac{5}{s^2}$$

Taking Laplace Inverse Transform of both sides

$$c(t) = 5.t.u(t)$$

• Proportional Integral Link:-

Proportional integral link's transfer function: $\frac{U_o(s)}{U_i(s)} = K + \frac{1}{TS}$.

Proportional integral link's diamonds chart, step response chart and simulation circuit chart are showed as following Figure 1.3.1, Figure 1.3.2 and Figure 1.3.3.

$K = \frac{R_2}{R_0 + R_1}$, $T = (R_0 + R_1)C_1$, parameter in the experiment $R_0=100K$ adjustable, $R_1=100K$,

$R_2=200K$, $C_1=1\mu F$, $R=10K$ or $100K$.

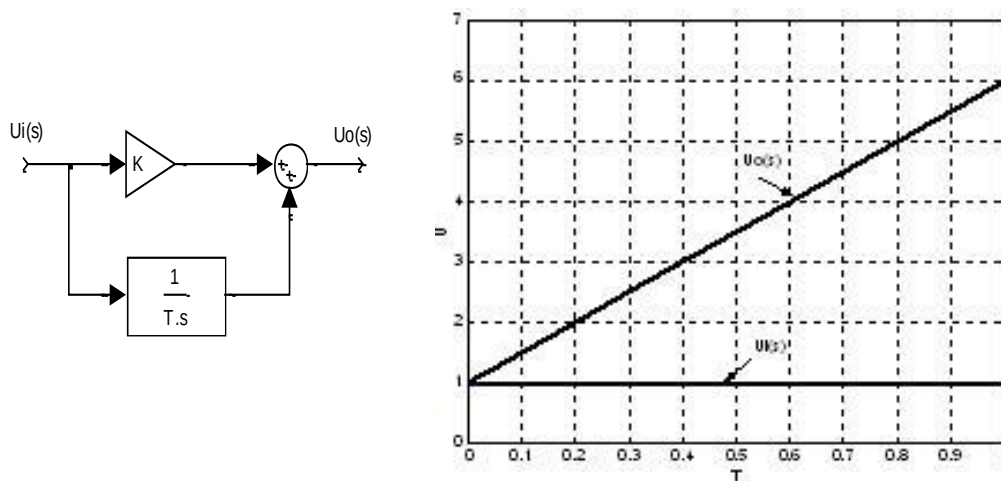


Figure 1.3.1

Figure 1.3.2

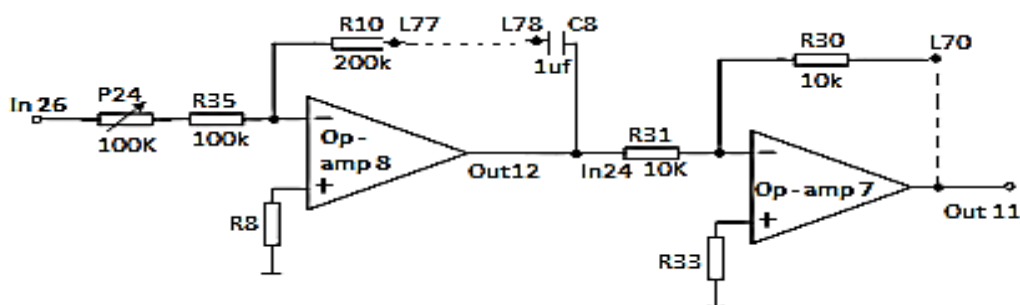


Figure 1.3.3

$$G(s) = K + \frac{1}{TS}$$

$$K = \frac{R_2}{R_0 + R_1} = \frac{200K}{200K} = 1$$

$$T = (R_0 + R_1)C_1 = (100K + 100K)10^{-6} = 0.2$$

$$\therefore G(s) = \frac{c(s)}{R(s)} = 1 + \frac{1}{0.2s}$$

$$\therefore c(s) = \frac{1}{s} + \frac{5}{s^2}$$

Taking Laplace Inverse Transform of both sides

$$c(t) = u(t) + 5.t.u(t)$$

- **Proportional Differential Link:-**

Proportional differential link's transfer function: $\frac{U_o(s)}{U_i(s)} = K(1 + TS)$.

Proportional differential link's diamonds chart, step response chart and simulation circuit chart are showed as following Figure 1.4.1, Figure 1.4.2 and Figure 1.4.3.

$$K = \frac{R_2 + R_3}{R_0 + R_1}, T = \frac{R_2 R_3}{(R_2 + R_3)} C_1, \text{ parameter in the experiment } R_0=100K \text{ adjustable, } R_1=10K,$$

$R_2=10K, R_3=10K, R_4=200\Omega, C_1=1\mu F, R=10 K \text{ or } 100K.$

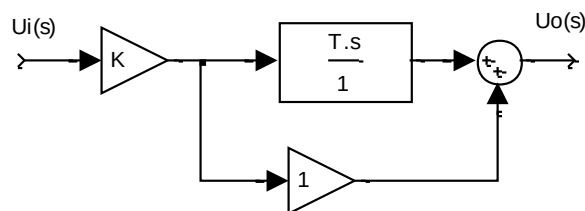


Figure 1.4.1

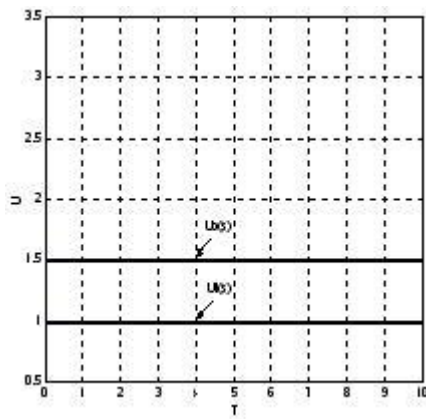


Figure 1.4.2a

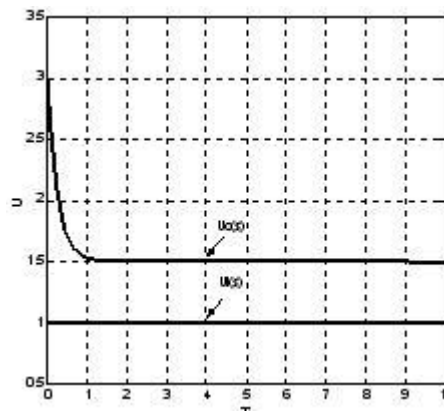


Figure 1.4.2b

Ideal and practical proportional differential link's step response chart showed as Figure 1.4.2a and Figure 1.4.2b.

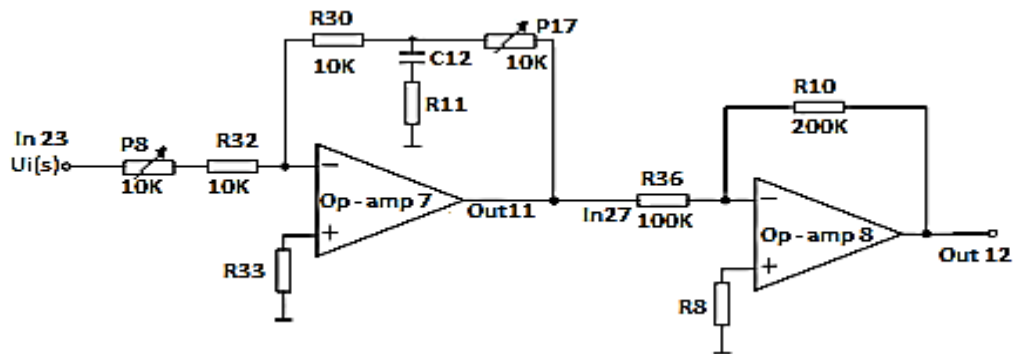


Figure 1.4.3

$$G(s) = K(1 + Ts)$$

$$K = \frac{R_2 + R_3}{R_0 + R_1} = \frac{10K + 10K}{10K + 10K} = 1$$

$$t = \frac{R_2 R_3}{(R_2 + R_3)} C_1 = \frac{10K \times 10K}{10K + 10K} \times 10^{-6} = 0.005$$

$$\therefore G(s) = 1(1 + 0.005s)$$

$$C(s) = \frac{1}{s} + 0.005$$

Taking Laplace Inverse Transform of both sides

$$c(t) = u(t) + 0.005\delta(t)$$

- **Inertia Link:-**

Inertia link's transfer function: $\frac{U_o(s)}{U_i(s)} = \frac{K}{TS+1}$.

Inertia link's diamonds chart, step response chart and simulation circuit chart are showed as following Figure 1.5.1, Figure 1.5.2 and Figure 1.5.3.

$K = \frac{R_2}{R_0 + R_1}$, $T = R_2 C_1$, parameter in the experiment $R_0=100K$ adjustable, $R_1=100K$,

$R_2=200K$, $C_1=1\mu F$, $R=10 K$ or $100K$.

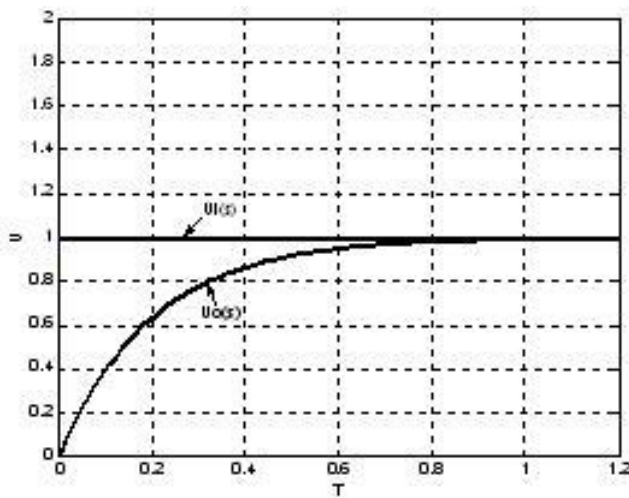


Figure 1.5.1

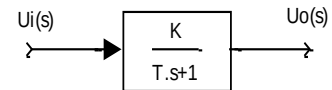


Figure 1.5.2

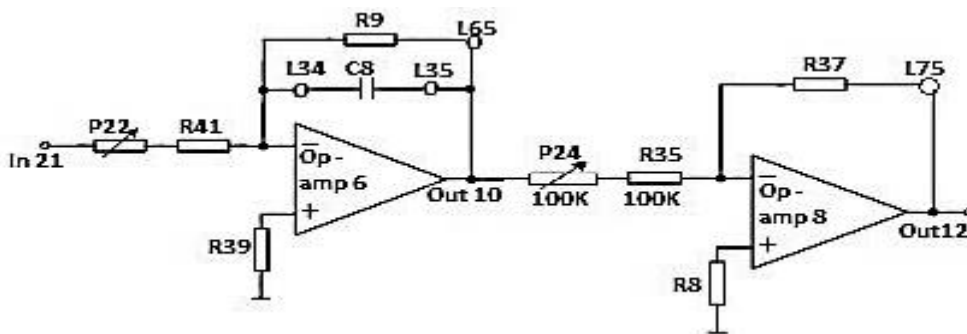


Figure 1.5.3

$$G(s) = \frac{K}{Ts + 1}$$

$$K = \frac{R_2}{R_0 + R_1} = \frac{200k}{200k} = 1$$

$$T = R_0 C_1 = 200k \times 10^{-6} = 0.2$$

$$\therefore G(s) = \frac{1}{0.2s + 1}$$

$$\therefore c(s) = \frac{1}{0.2s^2 + s}$$

$$= \frac{5}{s^2 + 5s} \dots\dots\dots [\text{Divide \& Multiply By 5}]$$

$$= \frac{A}{s} + \frac{B}{s + 5}$$

$$= \frac{1}{s} - \frac{1}{s + 5}$$

Taking Laplace Inverse Transform of both sides

$$c(t) = u(t) - e^{-5t}$$

- **PID Link:-**

PID link's transfer function: $\frac{U_o(s)}{U_i(s)} = K_p + \frac{1}{T_i S} + T_d S$

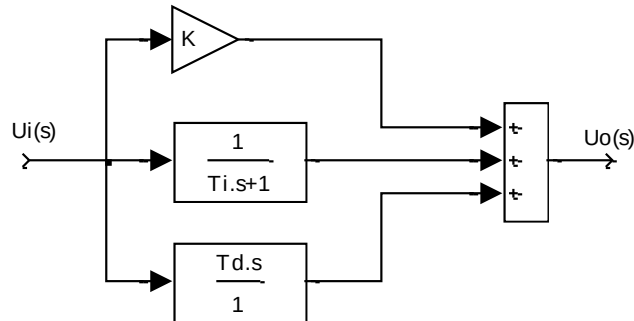


Figure 1.6.1

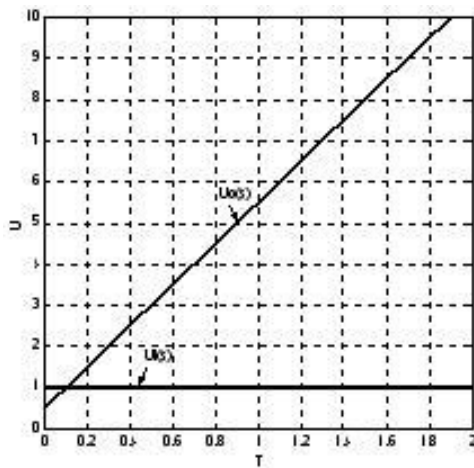


Figure 1.6.2a

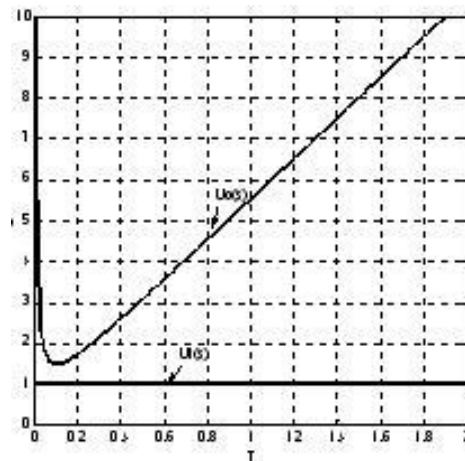


Figure 1.6.2b

PID link's diamonds chart, step response chart and simulation circuit chart are showed as following Figure 1.6.1, Figure 1.6.2 and Figure 1.6.3.

$$K_p = \frac{R_1}{R_0}, T_i = (R_1 + R_0)C_1, T_d = \frac{R_2 R_3}{(R_1 + R_0)} C_2, \text{ parameter in the experiment } R_0=100K$$

adjustable, $R_1=100K$, $R_2=100K$, $R_3=10K$, $R_4=1K$, $C_1=1\mu F$, $C_2=10\mu F$, $R=10K$ or $100K$.
Ideal and practical PID link's step response chart showed as Figure 1.6.2a and Figure 1.6.2b.

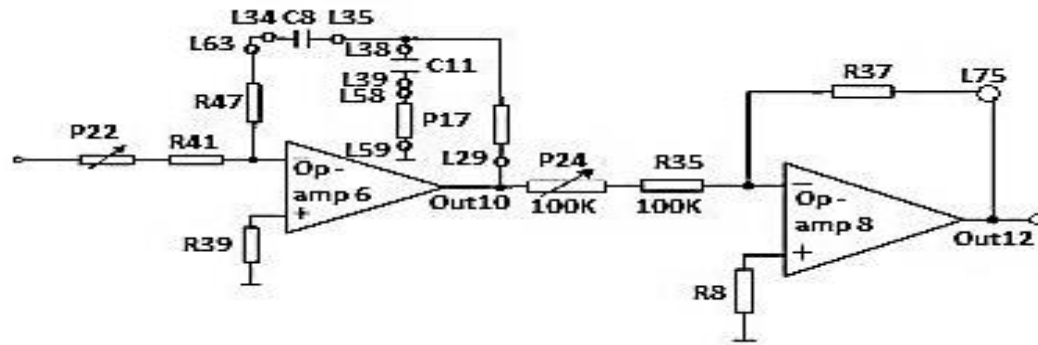


Figure 1.6.3

$$G(s) = Kp + \frac{1}{T_i s} + T_d s$$

$$Kp = \frac{R_1}{R_0} = \frac{100k}{100k} = 1$$

$$T_i = (R_1 + R_0)C_1 = 200k \times 10^{-6} = 0.2$$

$$T_d = \frac{R_2 R_3}{(R_1 + R_0)} C_2 = \frac{100k \times 10k}{(100k + 100k)} 10\mu F = 0.05$$

$$\therefore G(s) = 1 + \frac{1}{0.2s} + 0.05s$$

$$c(s) = \frac{1}{s} + \frac{5}{s^2} + 0.05$$

Taking Laplace Inverse Transform of both sides

$$c(t) = u(t) + 5t.u(t) + 0.05\delta(t)$$

PROCEDURES:-

- Get familiar with experiment apparatus, design and connect all kinds of typical link (proportional component, integral link, proportional integral link, proportional differential link, inertia link) simulation circuit according to experiment principle.
- According to experiment principle, proportional component of the experiment can use operational amplifier unit (I), (II), integral link can use operational amplifier unit (I), (II), proportional integral link can use operational amplifier unit (I), (II), proportional differential link can use operational amplifier unit (II), (III), inertia link can use operational amplifier unit (I), (II), PID link can use operational amplifier unit (III), (VI) and component unit.
- Finish all kinds of typical link simulation circuit's step characteristics test with experiment apparatus and research on the effect of parameter transformation on typical link step characteristics.

- Press switch SW2 and SW3 of power supply unit at one time, then press SW5, adjust the adjustable potentiometer P1, make $-12V$ — $+12V$ output voltage to $+1V$, forming unit step signal circuit, then press SW2 and SW3 at one time to turn off the power supply.
- Connect the circuit according to Figure 1.1.3 (proportional component), press the switch of operational amplifier unit in circuit.
- Connect circuit diagrams according to Figure 1.2.3(integral link), Figure 1.3.3 (proportional integral link), Figure 1.4.3 (proportional differential link), Figure 1.5.3 (inertia link), Figure 1.6.3 (PID link), repeat step (2), (3), (4).
- Analyze the results of experiment, finish experiment report.

RESULTS:-

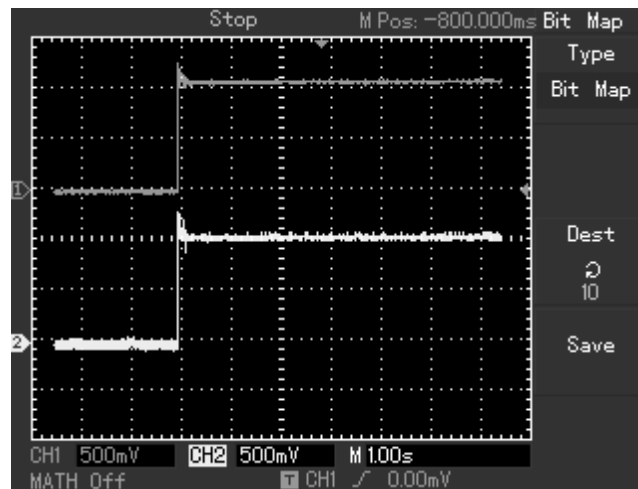
- Draw the simulation circuit diagrams of inertia link, integral link, proportional differential link, and also draw the response curves of inertia link, integral link, and proportional differential link.
- Figure out transfer functions of inertia link and integral link, according to step response curve, compare with the result which calculated according to circuit, analyze difference between them.

THINKING QUESTIONS:-

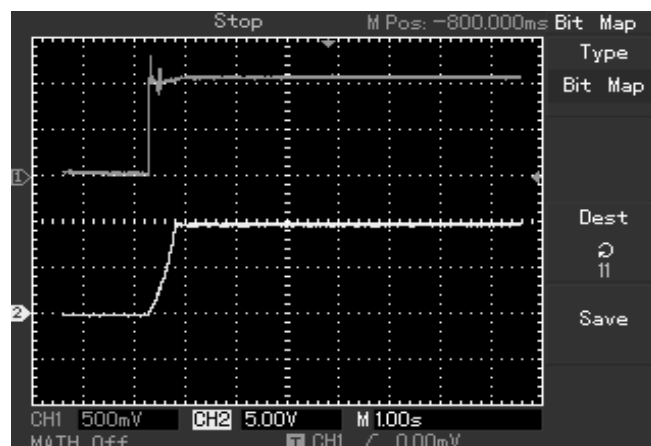
- When operational amplifier simulating typical links, under which two conditions of assumption can approximate, derive the transfer function?
- Under what condition, the inertia link can be approximate seen as integral link? Under what condition, it also can be approximate seen as proportional link?
- How to ascertain integral links and inertia link's time constants according to step response wave?

CAPTURED WAVEFORM:-

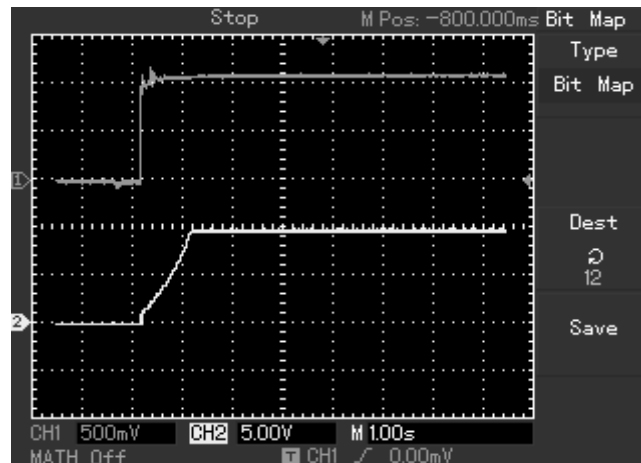
- 1) Step input response of Proportional link.



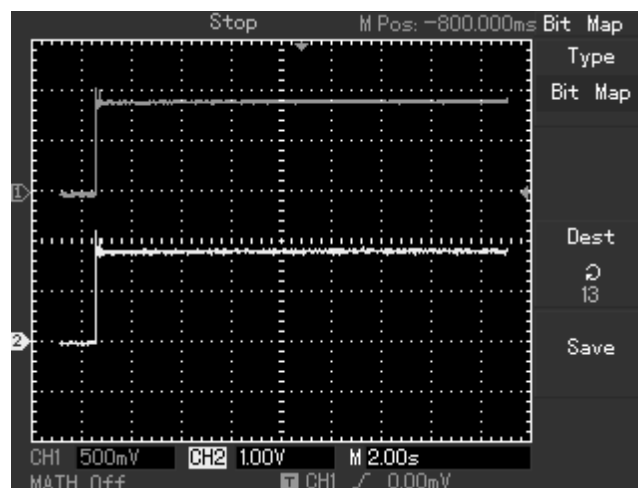
- 2) Step input response of Integral link.



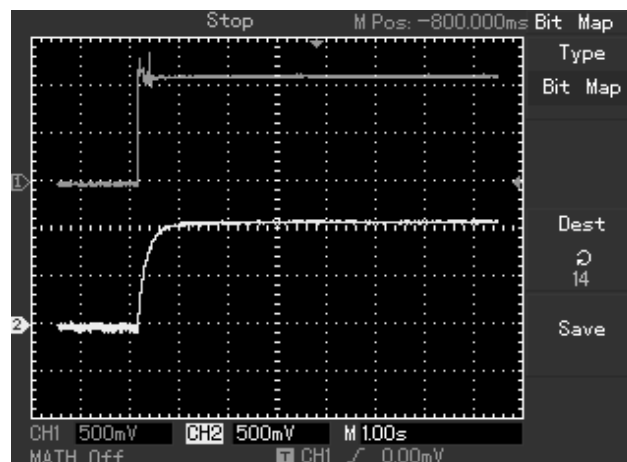
3) Step input response of Proportional Integral link.



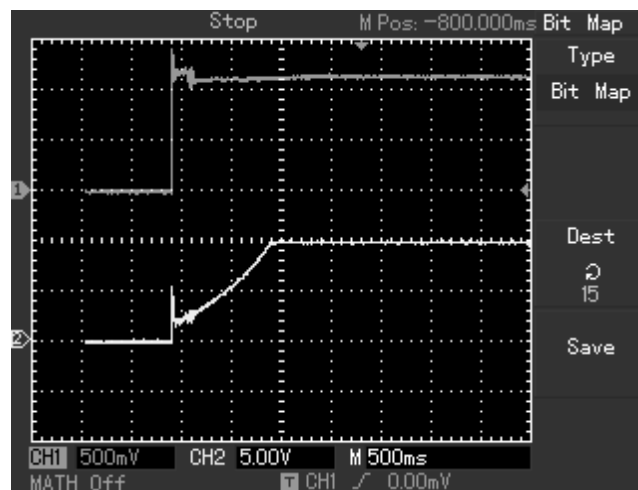
4) Step input response of Proportional Differential link.



5) Step input response of Inertia link.



6) Step input response of PID link.



EXPERIMENT

NO.2

EXPERIMENT NO:2

NAME:-

Step Response of Second-Order System and Stability Analysis.

OBJECTIVE:-

- Parameter to the influence of system dynamic performance and stability. Be familiar with the composition of second-order system.
- Study Step Response when second-order system are respectively working at the states of $\xi=1$, $0<\xi<1$, $\xi>1$, $\xi=0$.
- Learn to master testing method of dynamic performance index, and make research typical system.

APPARATUS:-

1. CL-PID :Op-Amp Application Trainer Kit.
2. Dual-trace low-frequency slow-scanning oscilloscope.
3. Digital Multi-meter.
4. Connecting Links.

THEORY:-

Experimental Contents

Observe Step Response of typical second-order system and measure system's Overshoot and adjusting time, and make research its parameter variety to the influence of system dynamic performance and stability.

Experimental Principles

Typical second-order system's block diagram is showed as Figure 2.1.

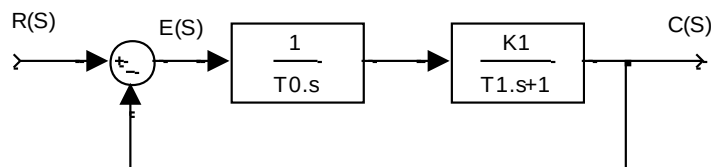


Figure 2.1

Its open-loop transmitting function is $G(S) = \frac{K}{S(T_1S + 1)}$, $K = \frac{K_1}{T_0}$ is open-loop gain.

Its close-loop transmitting function is $W(S) = \frac{\omega_n^2}{S^2 + 2\xi\omega_n S + \omega_n^2}$, among it, $\omega_n = \sqrt{\frac{K_1}{T_1 T_0}}$,

$$\xi = \frac{1}{2} \sqrt{\frac{T_0}{K_1 T_1}}.$$

Second-order system's analog circuit diagram is showed as Figure 2.2:

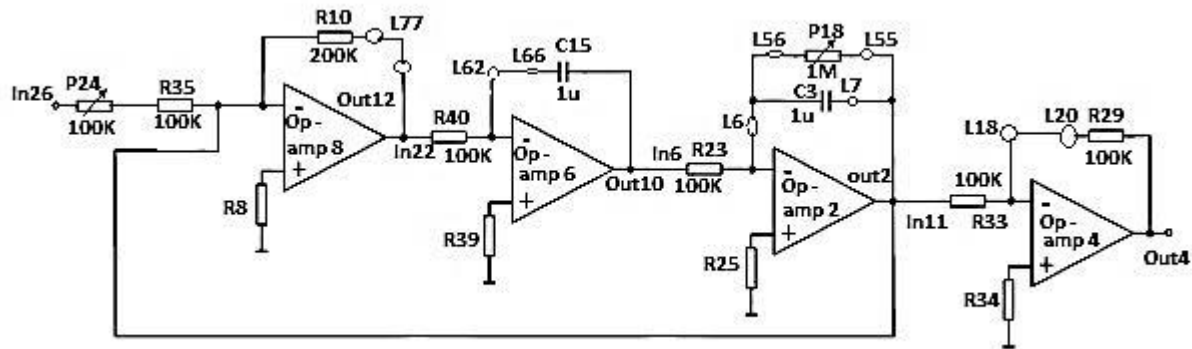


Figure 2.2

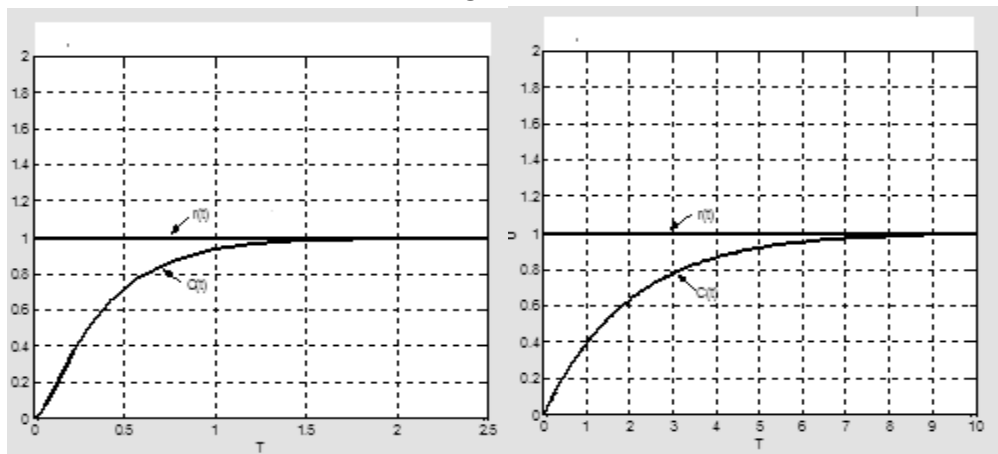


Figure 2.3.1

Figure 2.3.2

In the circuit, $T_0 = R_0 C_1$, $T_1 = R_x C_2$, $K_1 = \frac{R_x}{R_1}$.

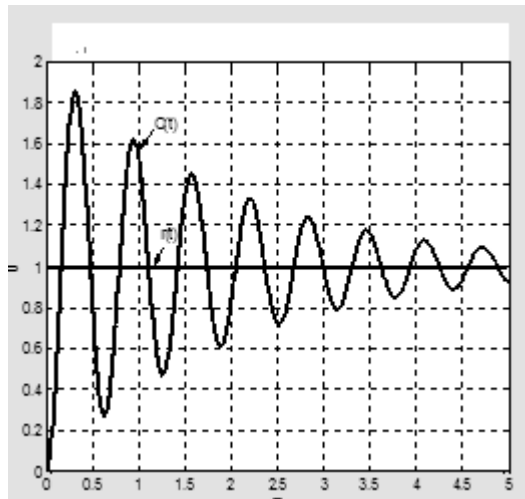


Figure 2.3.3

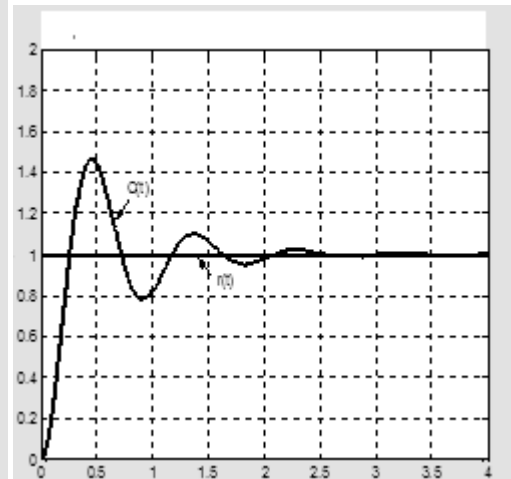


Figure 2.3.4

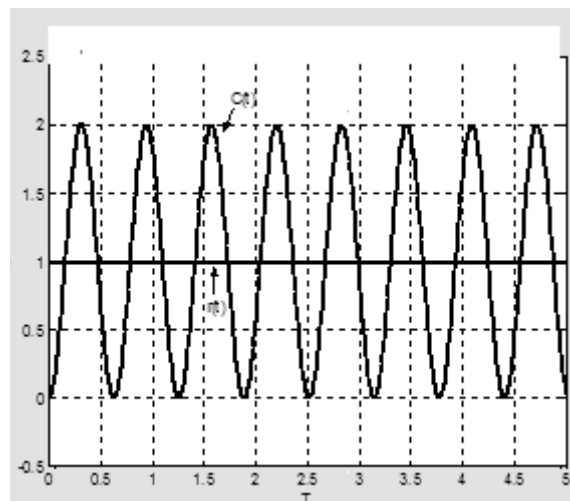


Figure 2.3.5

The second-order system's step response is showed as the above diagrams: Figure 2.3.1, 2.3.2, 2.3.3, 2.3.4 and 2.3.5 are respectively corresponding to step response curves when second-order at states of over-damping ($\xi > 1$), critical damping ($\xi = 1$), under-damped ($0 < \xi < 1$), damped oscillation with un-equivalent amplitude (ξ close to 0) and zero-damping ($\xi = 0$).

Change the value of component parameter R_x , it could make research time-domain response in different parameter characteristics. When $R_x = 50K\Omega$, the second-order system is working at the state of critical damping state; when $R_x < 50K\Omega$, second-order system is working at the state of over-damped state; when $R_x > 50K\Omega$, second-order system is

working at the state of under-damped state; when R_x is un-equivalent-amplitude damping oscillation; when $\xi=0$, the damping of second-order is zero, therefore the output shows equivalent-amplitude oscillation (Because all leads have resistance values, various losses exist all the time; the practical system's damping ratio ξ is impossible zero).

1. Expression for ω_n & ξ

From Figure 2.1,

$$G(s) = \frac{1}{T_0 \cdot s} \times \frac{K_1}{T_1 s + 1} = \frac{k_1}{T_0 T_1 s^2 + T_0 s}$$

$$H(s) = 1$$

$$\therefore \frac{C(s)}{R(s)} = \frac{G(s)}{1 + G(s)H(s)} = \frac{\frac{K_1}{T_0 T_1 s^2 + T_0 s}}{1 + \frac{K_1}{T_0 T_1 s^2 + T_0 s} \times 1}$$

$$= \frac{K_1}{T_0 T_1 s^2 + T_0 s + K_1}$$

$$= \frac{\frac{K_1}{T_0 T_1}}{s^2 + \frac{s}{T_1} + \frac{K_1}{T_0 T_1}} \dots\dots\dots \text{Devide NR \& DR By } T_0 T_1$$

Compare the above equation by 2nd order system's transfer function (C.L.T.F)

$$\frac{C(s)}{R(s)} = \frac{\omega_n^2}{s^2 + 2\xi\omega_n s + \omega_n^2}$$

Then we get,

$$\omega_n^2 = \frac{K_1}{T_0 T_1} \quad \& \quad 2\xi\omega_n = \frac{1}{T_1}$$

$$\omega_n = \sqrt{\frac{K_1}{T_0 T_1}} \quad \& \quad \xi = \frac{1}{2} \sqrt{\frac{T_0}{K_1 T_1}}$$

From the circuit,

$$T_0 = R_0 C_1, T_1 = R_x C_2, K_1 = \frac{R_x}{R_1}$$

As we know that,

Peak Overshoot

$$M_p \% = e^{-\pi\xi / \sqrt{1-\xi^2}} \times 100$$

Settling Time

$$T_s = 4 \times \text{Time Constant} = \frac{4}{\xi \omega_n}$$

Observation Table:-

Sr. No	R _x	ξ	ω_n	THEROTICAL		PRACTICAL	
				M _p	T _s	M _p	T _s
1	50K	Critical					
2	R _x >50k	0< ξ <1 Under down					
3	R _x <50k	ξ >1 Over down					
4	R ₀ =0 R _x =....	ξ =0					

PROCEDURES:-

- Make use of experimental apparatus, according to experimental principle diagram design; connect analog circuit of second-order close-loop system composed by an integral link and inertia link. In this experiment, you could use adjustable resistances in operational unit (I), (II), (III), (V) as well as component unit.
 - Press down keyswitch SW2, SW3 in power unit, then press down SW5, adjust potentiometer P1 to make -12V–+12V output voltage is +1V and form unit step signal circuit, then press down SW2, SW3 again, then power off the supply.
 - Connect the circuit correctly according to Figure 2.2, press down key-switch used in operational unit.
- Adjust the adjustable resistance, and repeat the above procedures, observe step characteristic curve when system at states of overdamped, critical damping and underdamped, and record waveform results at various states. Short-circuit resistance R_1 in Figure 2.2 to make $R_1=0$, at the same time, readjust resistance R_x to make the link to obtain proper proportion coefficient and time constant, then the system enters into stable equivalent-amplitude oscillation state.
- Change parameter of second-order analog circuit, repeat the above procedures, and observe parameter variety to the influence of system dynamic performance.
- Analyze experimental results and complete experimental report.

RESULT:-

Draw second-order's analog diagram, and calculate expression of parameters ξ , ω_n .

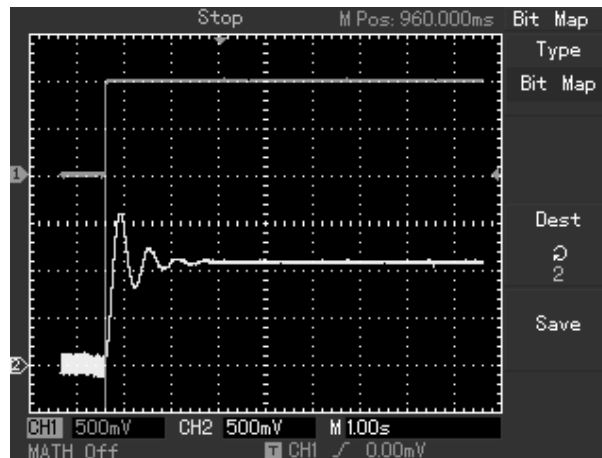
1. Calculate measurement values of M_p and t_s in different conditions of ξ and ω_n , observe results and make corresponding conclusion.
2. Draw response curve of second-order system, and calculate system's transmitting function by M_p and t_s values, and then compare them with the transmitting function calculated by analog circuit.

THINKING QUESTIONS:-

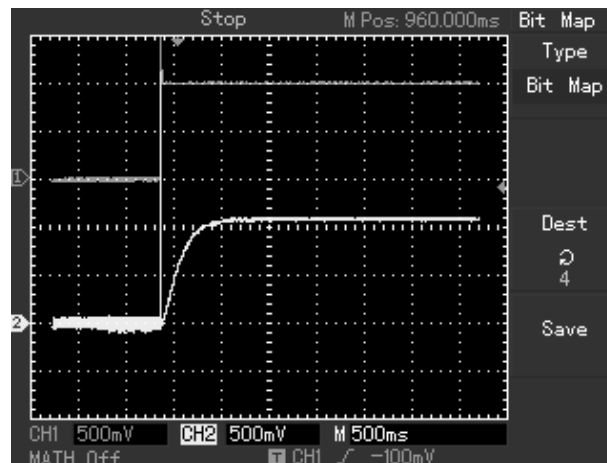
- If amplitude value of step input signal is too big, what results would be caused in experiment?
- How to ensure the system to realize negative feedback in experimental circuit? If there is an even operational amplifier in negative feedback loop, what feedback would be composed?
- If in analog experiment, the stable value of $C(t)$ is not equivalent to the amplitude value of step input function $r(t)$, what is the main reason?

CAPTURED WAVEFORM:-

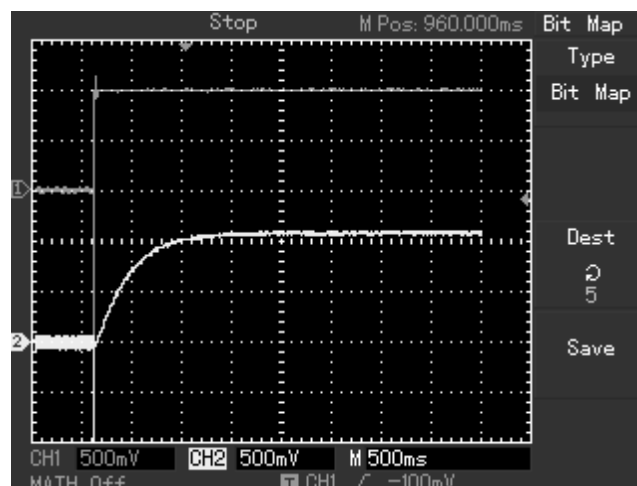
- 1) Step input response of second order system.



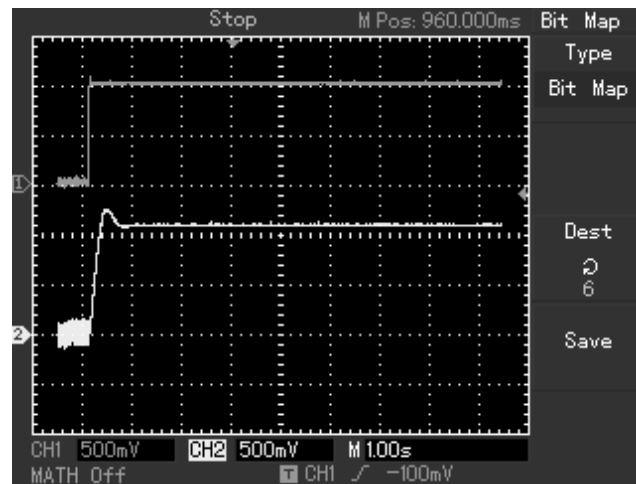
- 2) Step input response of second order system when $R_x = 50K$ (critical damping).



- 3) Step input response of second order system when $R_x < 50K$ (over-damped state).



4) Step input response of second order system when $R_x > 50K$ (under-damped state).



EXPERIMENT

NO.3

EXPERIMENT NO:3

NAME:-

Step Response of Third-order System and Stability Analysis.

OBJECTIVE:-

- Be familiar with the composition of three-order analog system.
- Study step response of three-order system, and observe its open-loop gain K to the influence of dynamic performance of three-order system.
- Learn to master testing methods of dynamic performance index and study typical system parameter to the influence of system dynamic performance and stability.

APPARATUS:-

- CL-PID:Op-Amp Application Trainer kit.
- Dual-trace low-frequency slow-scanning oscilloscope.
- Digital Multi-meter.
- Connecting Links.

THEORY:-

Experimental Contents

Observe step response of three-order system, measure system's overshoot and adjusting time, and study its parameter variety to the influence of system dynamic performance and stability.

Experimental Principles

Typical second-order system's block diagram is showed as Figure 3.1.

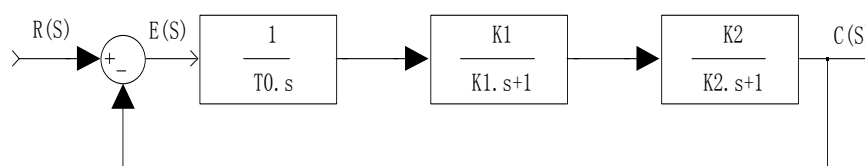


Figure 3.1

Its Open-Loop Transmission Function is $G(S) = \frac{K}{S(T_1S + 1)(T_2S + 1)}$, among it, $K = \frac{K_1K_2}{T_0}$.

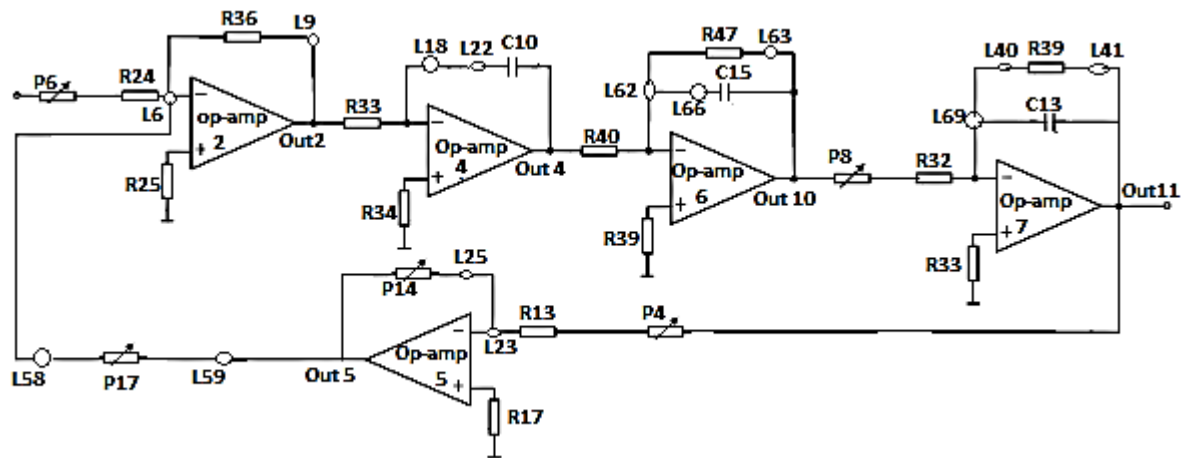


Figure 3.2

The Analog Circuit of the-three order system is shown as Figure 3.2.

The Open Loop Transmission Function of this system is $G(S)H(S) = \frac{K}{S(0.1S + 1)(0.5S + 1)}$,

among it, $K = \frac{500}{R_x}$, the unit of R_x is $K\Omega$.

The system characteristic equation is $s^3 + 12s^2 + 20s + 20K = 0$, which is obtained according to Routh Criterion.

System Stability $0 < K < 12$

System Critical Stability $K = 12$

System Instability: $K > 12$

According to K , R_x could be calculated; changing R_x , it could realize three kinds of experiments.

The step response diagram of this system is showed as following. Figure 3.3.1, Figure 3.3.2 and Figure 3.3.3 are corresponding to such three conditions as System Stability, System Critical Stability as well as System Instability.

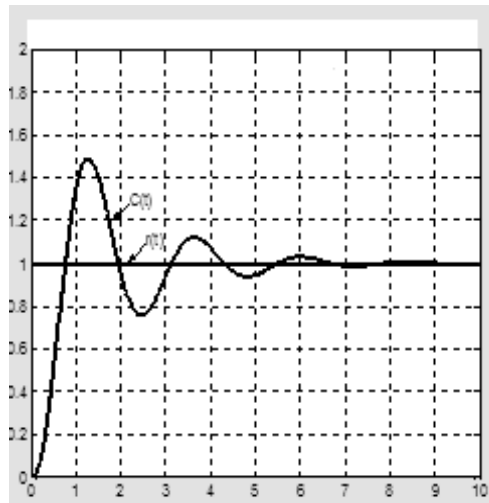


Figure 3.3.1

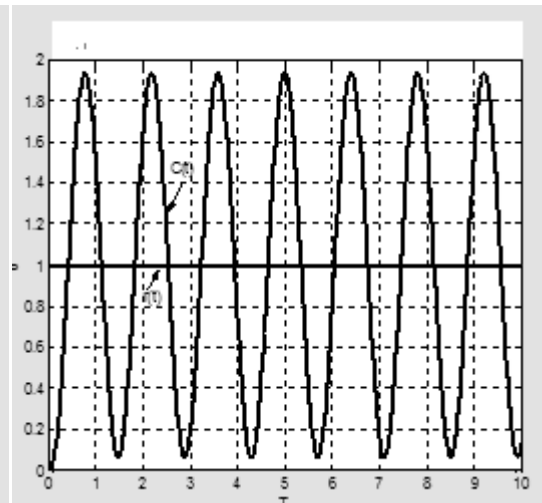


Figure 3.3.2

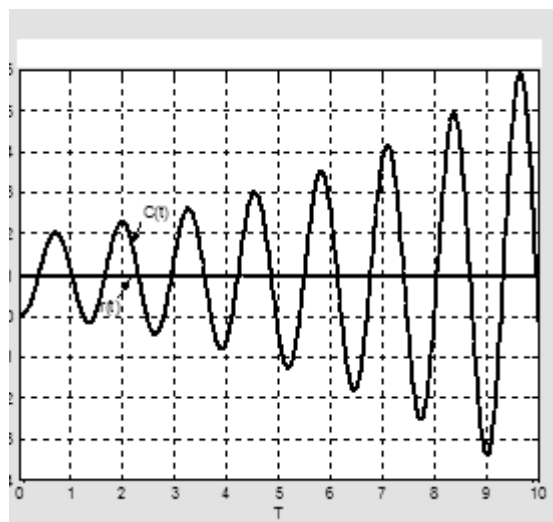


Figure 3.3.3

PROCEDURES:-

- By using experimental apparatuses, design and connect an analog circuit of three-order closed-loop system composed by an integral link and two inertia links according to experimental principle. In this experiment, the capacitance and adjustable resistance in Operational Amplifier Unit (I), (II), (III), (V), (VI) and Component Unit could be used.
- Simultaneously press down switches SW2, SW3 in power unit, then press down SW5, adjust adjustable potentiometer P1 to make the output voltage of -12V--+12V as +1V, in this way, the unit step signal circuit could be formed. Then press down SW2, SW3 again to switch off the power supply.
- Connect circuit well according to Figure 3.2, and press down the switch of operational unit used in circuit.
- Connect the connected input port of analog circuit with IN10, output port with

oscilloscope with the lead.

- Simultaneously press down switches SW2, SW3, observe step characteristics of this three-order system analog circuit with oscilloscope, measure overshoot and adjusting time of the response by experiment, and then record results.
- Adjust the adjustable circuit, repeat the above procedures, and observe step characteristic curves in states of Instability, Critical Stability, and Stability, and record waveform results correspondingly.
- Change parameters of three-order system analog circuit, repeat the above procedures, and observe influence of parameter variation to system dynamic performance.
- Analyze experimental results and finish the experimental reports.
- If sine wave is not showing then rotate P19.
- Also rotate P8 for the proper signal.

EXPERIMENTAL ANALYSIS:-

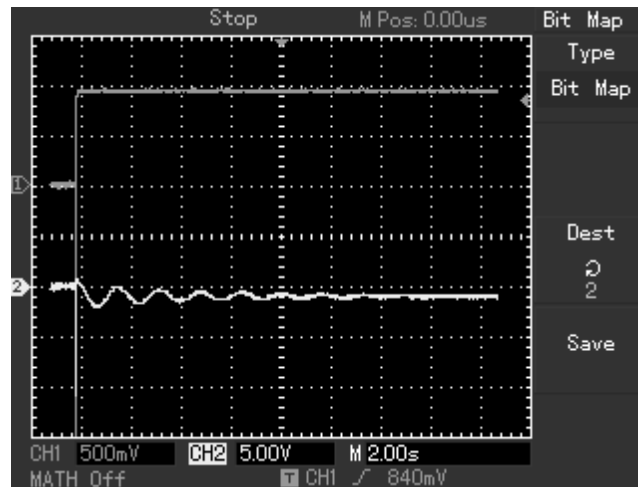
- Draw typical analog circuit diagram of the three-order system.
- Draw waveforms diagram of (amplitude reduction) in Stability state and (amplitude increase) in Instability state for the three-order system.
- Calculate the critical amplifying coefficient of the three-order system, and compare it with the one measured in analog circuit.

THINKING QUESTIONS:-

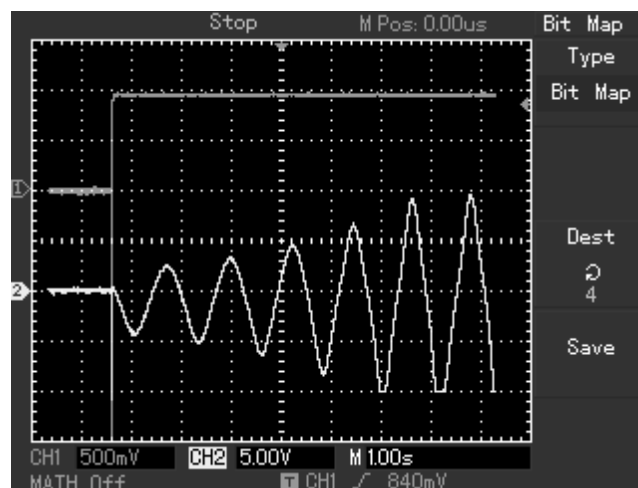
- In order to promote the system to work stably, should the open-loop gain set smaller or larger properly?
- As for the small inertia link and big inertia link in system, which has larger influence to system Stability, and why?
- Try to explain in the three-order system experiment, why did the output appear equivalent-amplitude oscilloscope with top-cut?
- If the stable value of $C(t)$ is not equivalent to amplitude value of step input function $r(t)$, what are the main reasons?

CAPTURED WAVEFORM:-

- 1) Step input of third order system ($0 < K < 12$) System Stability.



- 2) Step input of third order system ($K > 12$) System Instability.



EXPERIMENT

NO.4

EXPERIMENT NO:4

NAME:-

Dynamic Characteristic of PID Controller.

OBJECTIVE:-

- Be familiar about analog circuit of such three controllers as PI, PD and PID.
- By experiment, further know about step response characteristic of such three controllers as PI, PD and PID and influences of relative parameters to their performances.

APPARATUS:-

- CL-PID: Op-Amp Application Trainer KIT.
- Dual-trace low-frequency slow-scanning oscilloscope.
- Digital Multi-meter.
- Connecting links.

THEORY:-

Experimental Contents

- Complete dynamic characteristic experiment of PI controller.
- Complete dynamic characteristic experiment of PD controller.
- Complete dynamic characteristic experiment of PID controller.

Experimental Principles

PI, PD and PID controllers are active correction devices widely applied in industrial control system. Among it, PD is advanced correction device, which is applied in such occasion that the stable performance has been satisfied while the dynamic performance is quite poor. PI is the lag correction device, which could change system's stable performance. PID is lag---advance correction device, which has advantages of PI and PD.

• PD Controller:-

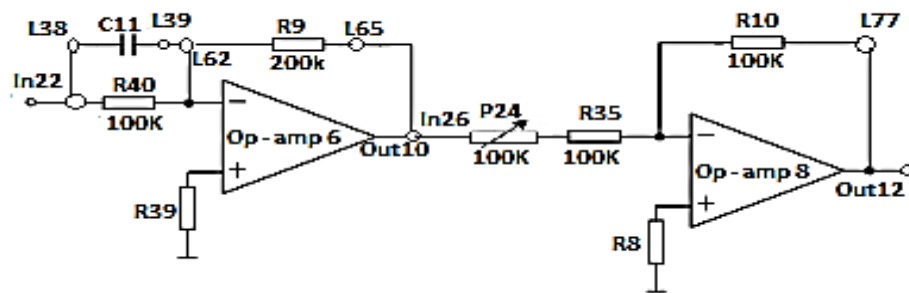


Figure 4.1

Figure 5.1 is the diagram of PD controller, its transmission function is $G(S) = K_p(T_d S + 1)$,

among it, $K_p = \frac{R_2}{R_1}$, $T_d = R_1 C_1$.

Choose $R_1=100K$, $R_2=200K$, $C=1\mu F$, C11(use electrolytic capacitor) the transmission function is $G(S) = 2(0.1S + 1)$. The step response diagram is shown as Figure 4.2.

PD Controlling Law: its function is to response the variation tendency of input signal, generate effective early emended signal to increase system's damping degree, thus to change system's stability.

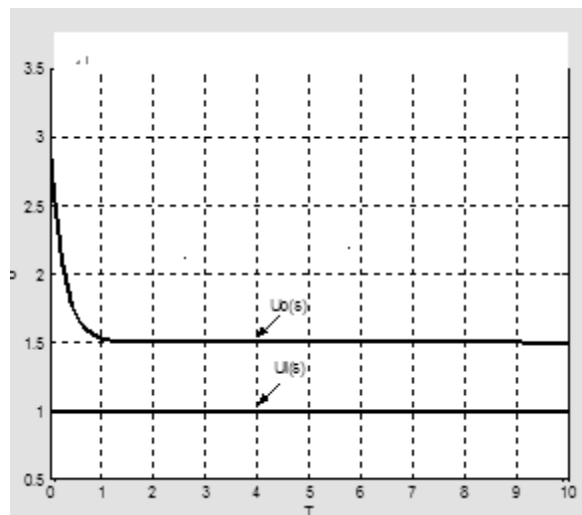


Figure 4.1

$$G(s) = K_p (T_d s + 1)$$

$$K_p = \frac{R_2}{R_1} = \frac{200k}{100k} = 2$$

$$T_d = R_1 C_1 = 100k \times 10^{-6} = 0.1$$

$$\therefore G(s) = 2(0.1s + 1)$$

$$= 0.2s + 2$$

$$\therefore C(s) = 0.2 + \frac{2}{s}$$

Taking Laplace Inverse Transform of both sides

$$\therefore C(t) = 0.2\delta(t) + 2u(t)$$

• PI Controller:-

Figure 5.3 is Diagram of PI controller, its transmission function is: $G(S) = K_p \left(1 + \frac{1}{T_i S}\right)$,

among it, $K_p = \frac{R_2}{R_1}$, $T_i = R_2 C_2$.

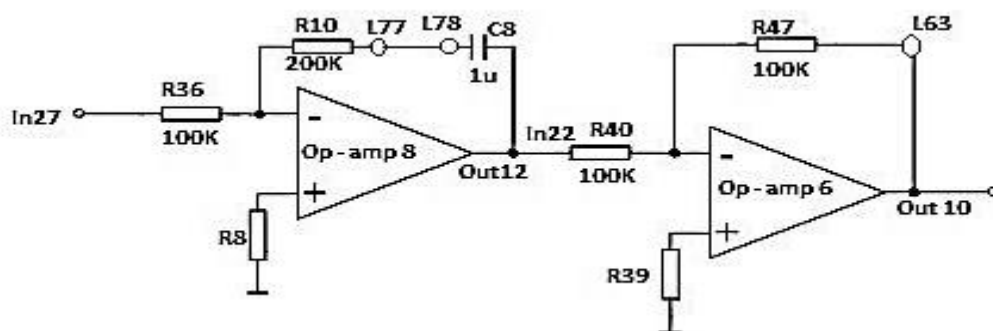


Figure 4.3

Choose $R_1=200K$, $R_2=200K$, $C_2=1\mu F$, its transmission function is: $G(S) = 1 + \frac{1}{0.2S}$, its step response is shown as Figure 4.4.

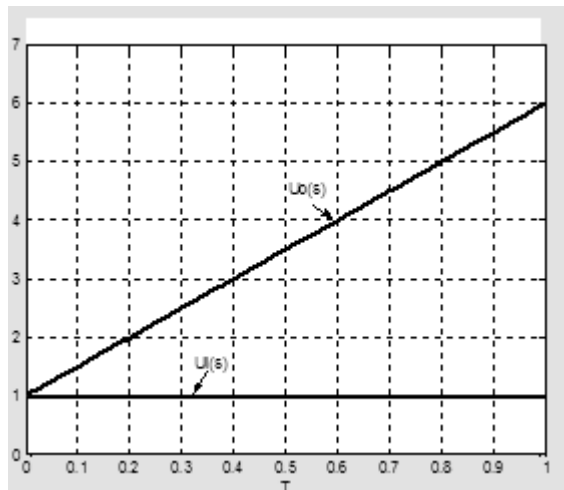


Figure 4.4

PI Control Law: its function is equivalent to add an open-loop pole and open-loop zero. The pole located at the origin could improve the stable performance of system while the increased zero could ease the adverse effects.

$$G(s) = K_p \left(1 + \frac{1}{T_i s}\right)$$

$$K_p = \frac{R_2}{R_1} = \frac{200k}{200k} = 1$$

$$T_i = R_2 C_2 = 200k \times 10^{-6} = 0.2$$

$$\therefore G(s) = 1 \left(1 + \frac{1}{0.2s}\right)$$

$$\therefore C(s) = \frac{1}{s} + \frac{5}{s^2}$$

Taking Laplace Inverse Transform of both sides

$$\therefore C(t) = u(t) + 5t.u(t)$$

- **PID Controller:-**

Figure 5.5 is diagram of PID controller, its transmission function is $G(S) = K_p \frac{(1+TS)(1+\tau S)}{TS}$,

among it $K_p = \frac{R_2}{R_1}$, $T = R_2 C_2$, $\tau = R_1 C_1$.

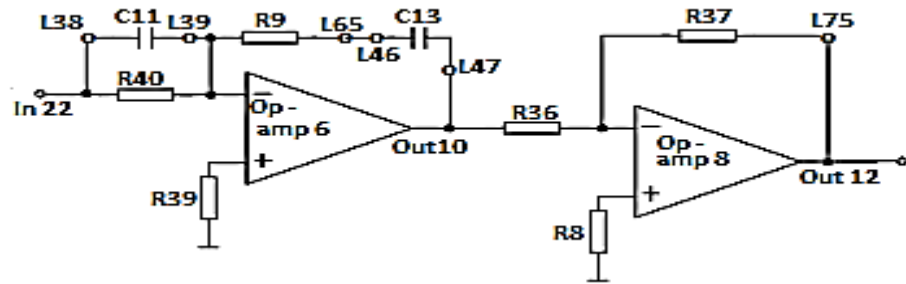


Figure 4.5

Choose $R_1=100K$, $R_2=200K$, $C_1=1\mu F$, $C_2=10\mu F$, C_{11} (use electrolytic capacitor) then the transmission function is: $G(S) = 2(1.05 + 0.1S + \frac{1}{2S})$. The step response is shown as Figure 4.6.

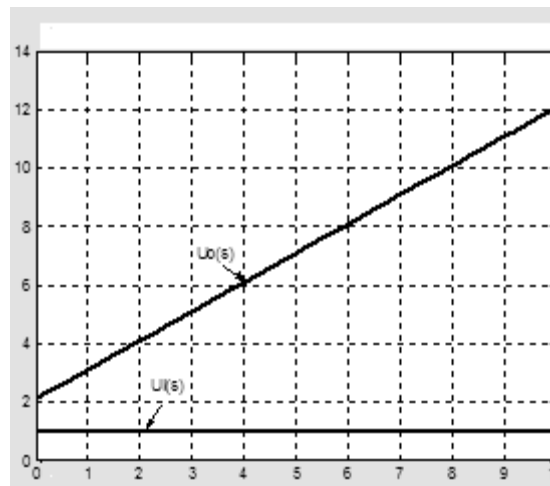


Figure 4.6

PID Control Law: compared with PI controller, besides the advantage of improving system stable performance, its function has greater advantage of improving system's dynamic performance because the system adds a negative real part zero.

$$G(s) = K_p \frac{(1+T.s)(1+\tau.s)}{T.s}$$

$$K_p = \frac{R_2}{R_1} = \frac{200k}{100k} = 2$$

$$T = R_2 C_2 = 200k \times 10\mu F = 2$$

$$\tau = R_1 C_1 = 100k \times 10^{-6} = 0.1$$

$$\begin{aligned}
 G(s) &= 2 \frac{(1+2s)(1+0.1s)}{2s} \\
 &= \frac{1+0.1s+2s+0.2s^2}{s} \\
 &= \frac{1+2.1s+0.2s^2}{s}
 \end{aligned}$$

$$\therefore C(s) = \frac{1}{s^2} + \frac{2.1s}{s} + 0.2$$

Taking Laplace Inverse Transform of both sides

$$\therefore C(t) = r(t) + 2.1u(t) + 0.2\delta(t)$$

PROCEDURES:-

- Take advantage of experimental apparatuses, according to experimental principle, design and connect PD controller of analog circuit. Observe PD controller's output waveform with oscilloscope, and record output waveform. PD controller could use capacitances in operational unit (I), (II) and components.
 - Simultaneously press down switches SW2, SW3 in power unit, then press down SW5; adjust potentiometer P1 to make output voltage -12V+12V as +1V to form unit step signal circuit; press down SW2, SW3 again, and then close the power supply.
 - Connect the circuit according to Figure 4.1 (PD Controller), press down switch of operational unit used in circuit.
- Respectively vary values of R_1 , R_2 and C , observe their output waveforms and record them.
- Take advantage of experimental apparatuses, according to experimental principle, design and connect analog circuit of PI Controller, PI Controller could use operational unit (I), (II). Repeat the above procedures (1), (2), (3), (4), (5), observe output waveform of PI Controller, and record results.
- Take advantage of experimental apparatuses, according to experimental principle, design and connect analog circuit of PID Controller; PID Controller could use capacitances in operational unit (I), (II), and components. Repeat the above procedures (1), (2), (3), (4), (5), observe output waveform of PID Controller, and record results.
- Analyze experimental results, and complete experimental reports.

RESULTS:-

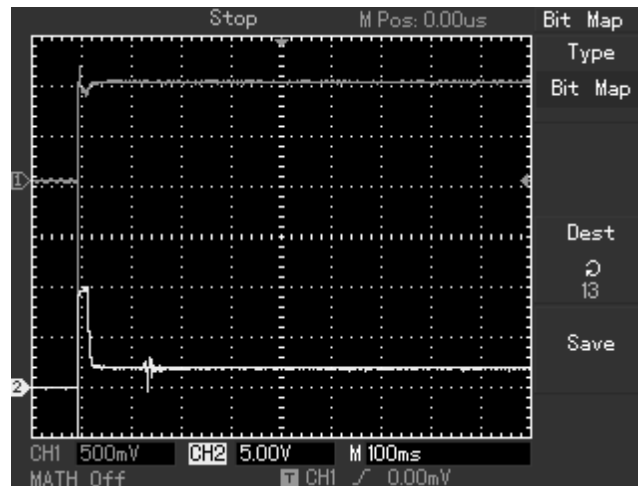
- Draw Diagram of such three Controllers as PI, PD and PID.
- According to transmission function of three controllers, draw theoretic output waveform diagram under the function of unit step signal for each controller.
- According to experimental results, draw unit step response curve of three controllers, and compare them with output waveforms that are calculated theoretically.
- Analyze influences of parameter to three controllers' performance.

THINKING QUESTIONS:-

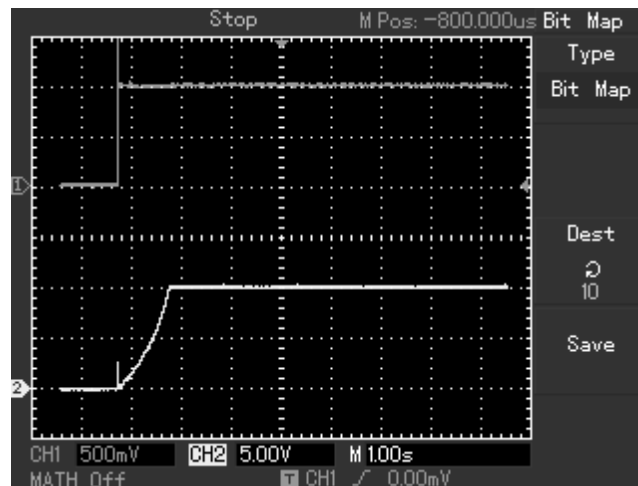
- What occasions is PD and PI applied? What are their advantages and disadvantages?
- What are advantages of PID Controllers?
- What are differences between output waveforms of PD and PID obtained by experiments and their theoretic waveforms?

WAVEFORM:-

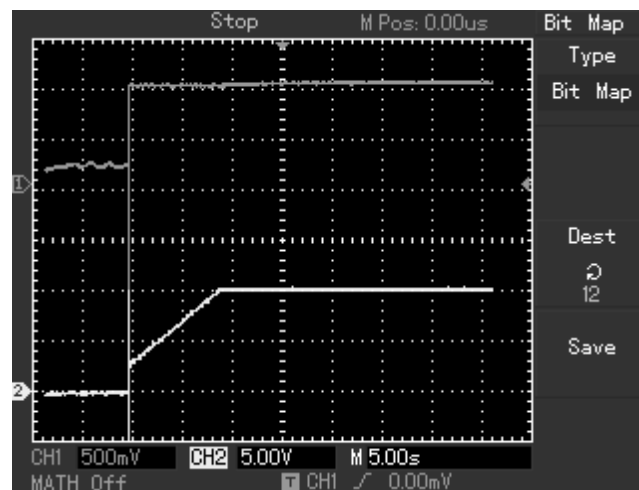
1) Dynamic Characteristics of PD Controller:



2) Dynamic Characteristics of PI Controller:



3) Dynamic Characteristics of PID Controller:



EXPERIMENT

NO.5

EXPERIMENT NO:5

NAME:-

Linear System Serial Connection Calibration Circuit Analog Experiment.

OBJECTIVE:-

- To be familiar with the impact of serial connection calibration devices to the stability and dynamic characteristics of the system.
- To master the design measures and parameter debugging technologies of the serial connection calibration devices.

APPARATUS:-

1. CL-PID:Op-Amp Application Trainer Kit.
2. Dual-Trace Low-Frequency Slow-Scan Oscilloscope.
3. Digital Multi-meter.
4. Connecting Links.

THEORY:-

Experimental Contents

- To observe stability and dynamic characteristics of the under calibration system.
- Design serial connection calibration devices according to dynamic characteristics.
- Observe the system stability and dynamic characteristics after equipped with serial connection calibration device, and observe the influence of changes of calibration device parameter on system performance.

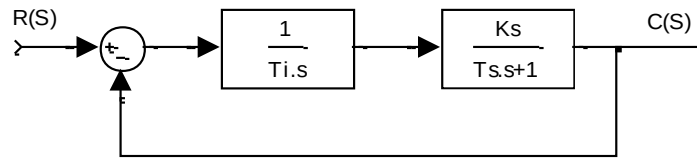
Experimental Principles

When the system open-loop gains meet its stability performance requirements, its dynamic performance is not satisfactory or even cause instability. Thus requires to serial connect a calibration device not only keep the system open-loop gains unchanged, but also make the system dynamic performance meet the requirements.

The open-loop transfer function of second-order system is:

$$W(S) = \frac{\omega_n^2}{S(S + 2\xi\omega_n)} = \frac{\frac{\omega_n^2}{(2\xi\omega)^2}}{S(\frac{S}{2\xi\omega} + 1)} \quad (6-1)$$

Its system block diagram is shown in Figure 6.1 below:

**Figure 5.1**

It's open-loop transfer function is:

$$W(S) = \frac{K_1}{\tau S(TS + 1)} = \frac{K}{S(TS + 1)} \quad (6-2)$$

Where $K = \frac{K_1}{\tau}$, compare (6-1) with (6-2), then get

$$T = \frac{1}{2\xi\omega_n} \quad (6-3)$$

$$\frac{K_1}{\tau} = \frac{\omega_n^2}{2\xi\omega_n} = \frac{\omega_n}{2\xi} \quad (6-4)$$

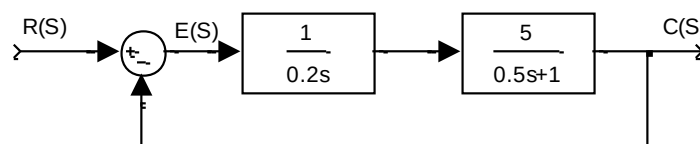
If require $\xi = \frac{\sqrt{2}}{2}$, then $T = \frac{1}{2\omega_n/\sqrt{2}} = \frac{1}{\sqrt{2}\omega_n}$, $\frac{K_1}{\tau} = \frac{\omega_n}{2/\sqrt{2}} = \frac{\omega_n}{\sqrt{2}} = \frac{1}{2T}$

When $\xi = \frac{\sqrt{2}}{2}$, Second-order system in standard form gets the closed-loop transfer function as:

$$T(S) = \frac{\omega_n^2}{S^2 + \sqrt{2}\omega_n S + \omega_n^2}, \text{ take } \omega_n = \frac{\sqrt{2}}{2T} \text{ into the equation can get } T(S) = \frac{1}{2T^2 S^2 + 2TS + 1}.$$

This is the best closed-loop transfer function of second-order system.

1. In this experiment, the block diagram and analog circuits we use are second-order closed-loop systems without calibration. They are respectively shown in Figure 6.2 and Figure 6.3:

**Figure 5.2**

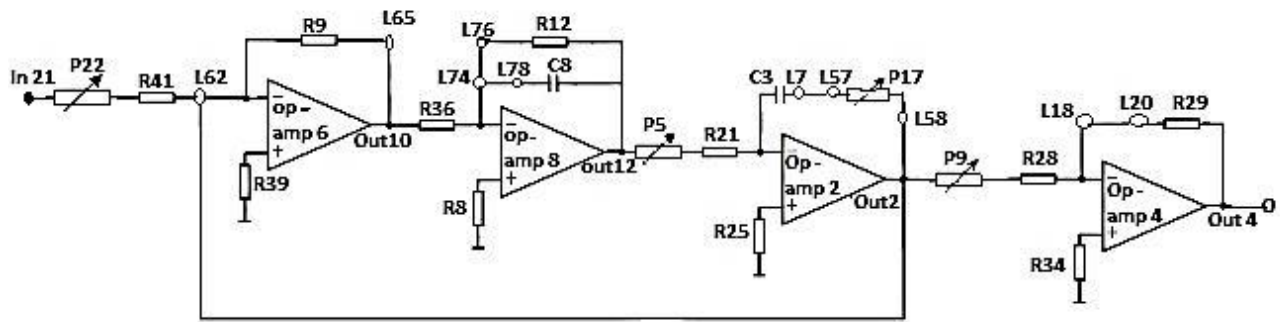


Figure 5.3

Its open-loop transfer function is $G(S)H(S) = \frac{5}{0.2S(0.5S + 1)}$

Its closed-loop transfer function is $W(S) = \frac{50}{S^2 + 2S + 50}$

system step response before serial connection calibration

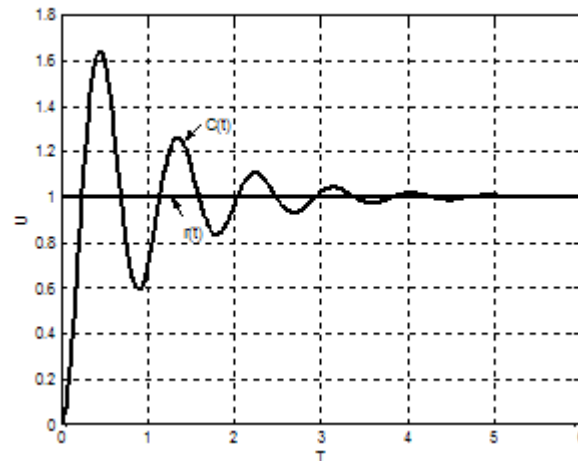


Figure 5.4

This step response curve of second-order system is shown as Figure 6.4. Its dynamic Index is: $M_p = 63\%$, $t_s = 4s$; Static error coefficient $K_v = 25 \text{ 1/s}$.

2. It requires the added serial connection calibration device system should meet following performance index:

- (1) overshoot $M_p \leq 20\%$
- (2) adjustment time (transition time) $t_s \leq 1s$
- (3) system open-loop gain after calibration (static error coefficient) $K_v \geq 25 \text{ 1 / s}$

Based on system requirements after calibration can count out the system open-loop transfer

function: $G(S)H(S) = \frac{5}{0.2S(0.1S + 1)}$

It's proved by theory that the damping ratio of this second-order system $\xi = \frac{\sqrt{2}}{2} = 0.707$; the

step response overshoot $M_p = 4.3\%$; adjust time is $0.8s$;
phase margin $\gamma = 63^\circ$.

3. Serial connection calibration device design

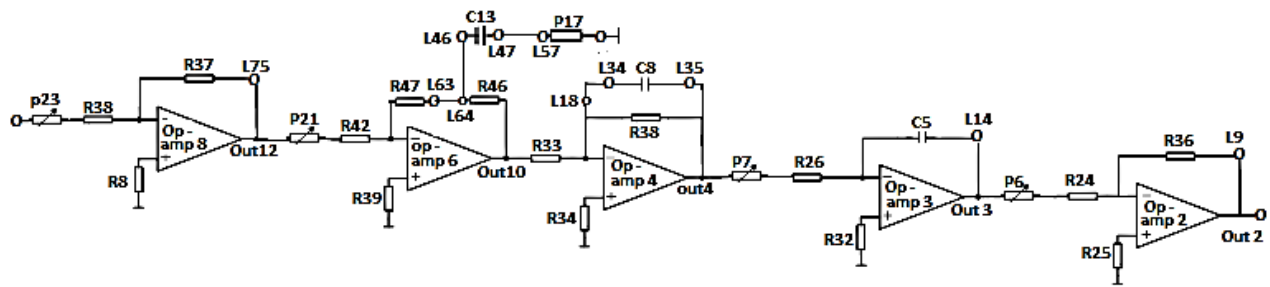


Figure 5.5

The desired open-loop transfer function of the system when divided by the second-order closed-loop system open-loop transfer function can get serial connection calibration device

$$\text{transfer function: } G_c(s) = \frac{0.5s + 1}{0.1s + 1}$$

From serial connection calibration device transfer function can design out its analog circuits. Figure 6.5 shows the analog circuits when added serial connection calibration device.

The parameters of serial connection calibration device circuit are set as: $R_1 = 200K\Omega$, $R_2 = R_3 = 100K\Omega$, $R_4 = 10K\Omega$, $C = 10\mu F$.

System step response curve after serial connection calibration is shown in Figure 6.6:

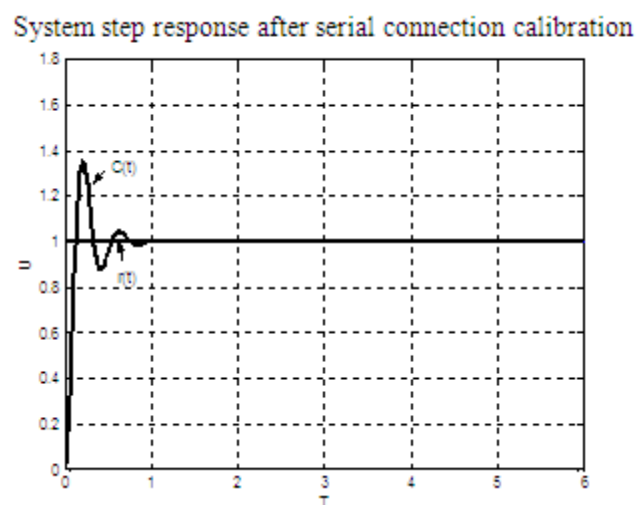


Figure 5.6

PROCEDURES:-

- According to experiment principles design and connect a second-order closed-loop system analog circuit with experiment APPARATUS to accomplish the stability and dynamic characteristics observation of this system. This experiment can use Op-amp units I, II, III, V.
- Press down Button Switches SW2 and SW3 of Power Unit at the same time, then press

- down SW5; adjust adjustable potentiometer P1 to make -12V-+12 V output voltage to be +1 V thus form unit step signal circuits, and then press down SW2 and SW3 again, then power off.
- Connect circuit according to Figure 5.3 (without calibration); press down the Button Switches used in Op-amp units.
 - When press down Button Switches SW2 and SW3 of Power Unit at the same time, observe step response curve of output end with oscilloscope and record down the waveforms result.
 - Use overshoot and adjust time g to design serial connection calibration device transfer function and analog circuits according to dynamic requirements.
 - According to experiment contents design and connect a second-order closed-loop system analog circuit after serial connection calibration device with experiment apparatuses to accomplish the stability and dynamic characteristics observation of this system. This experiment can use Op-amp units I, II, III, V, VI and the resistance and capacitance in component unit.
 - Press down Button Switches SW2 and SW3 of Power Unit at the same time, then press down SW5; adjust adjustable potentiometer P1 to make T006 (-12V-+12V) output voltage to be +1 V thus form unit step signal circuits, and then press down SW2 and SW3 again, then power off.
 - Connect circuit according to Figure 6.5 (after calibration); press down the Button Switches used in Op-amp units.
 - Connect the input ending of connected analog circuit with IN10 with wires; connect output ending of circuit with oscilloscope.
 - When press down Button Switches SW2 and SW3 at the same time, observe step response curve of output end with oscilloscope and record down the waveforms result.
 - Change serial connection calibration the parameters; debug second-order closed-loop system after calibration and make its performance index meet pre-determined requirements.
 - Analysis experiment results and finish experiment report.

RESULTS:-

- Draw the analog circuit and system structure diagram before calibration, and observe the system's step response curve.
- Count out σ and t_s before and after calibration according to reference standard.
- According to the counted σ and t_s to count system analog circuit diagram after calibration, and observe the step response curve.
- Draw the step response curves before and after calibration and analysis the calibration results.

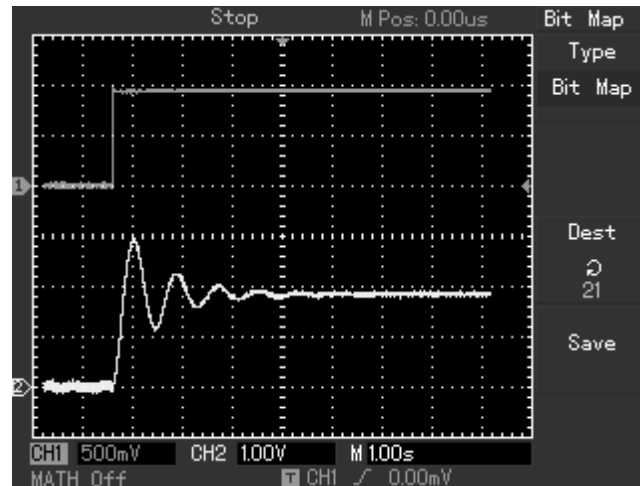
THINKING QUESTIONS:-

- Are there any other types of calibration devices in addition to advance calibration device? What are their characteristics? What are the differences? How to choose the types of calibration devices?

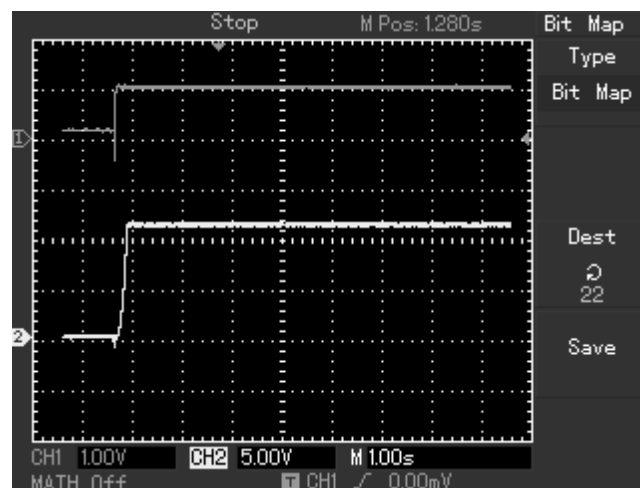
- If $G_c(S) = \frac{0.5S + 1}{0.05S + 1}$ in this experiment, can ω_n and ξ of the system after calibration be counted out?

CAPTURED WAVEFORM:-

- 1) System step response before serial connection calibration



- 2) System step response after serial connection calibration



EXPERIMENT

NO.6

EXPERIMENT NO:6

NAME:-

Non-Linear System Phase-Plane Method Circuit Analog Experiment.

OBJECTIVE:-

- To master non-linear system analog methods and learn to analyze non-linear system with phase plane method.
- To be familiar with the research on non-linear system circuit analog and master its analog research methods.

APPARATUS:-

- CL-PID:PID Control Circuit Trainer KIT.
- Dual-Trace Low-Frequency Slow-Scan Oscilloscope.
- Digital Multi-meter.
- Connecting Links.

THEORY:-

Experimental Contents

- To complete the experiment of relay-type non-linear closed-loop system without calibration.
- To complete the experiment of relay-type non-linear closed-loop system with velocity negative feedback.
- To complete the experiment of saturation-type non-linear closed-loop system.

Experimental Principles

Phase plane method is an effective method to analyze first-order and second-order nonlinear system. Through the drawn phase track we can directly know the system operating situation.

• **Relay-type Non-Linear Closed-Loop System without Calibration.**

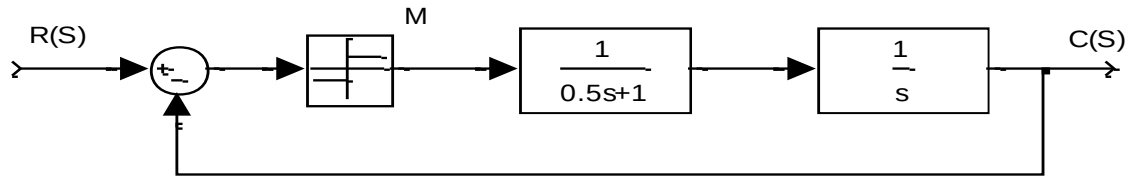


Figure 6.1

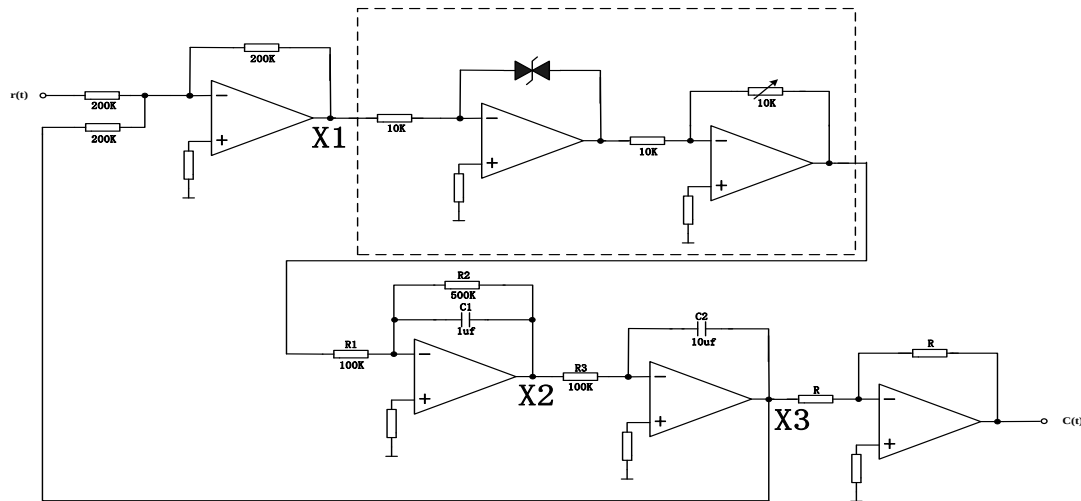


Figure 6.2

The principle block diagram of relay-type non-linear closed-loop system without calibration is shown as in Figure 8.1.

Its analog circuit is shown as in Figure 6.2.

The system shown in Figure 6.2 can be described by the following equations:

$$T\ddot{e} + \dot{e} - KM = 0 \quad e > 0$$

$$T\ddot{e} + \dot{e} + KM = 0 \quad e < 0 \quad (6-1)$$

Where T is time constant ($T = 0.5$); K is part of the open-loop gain of the linear part; M is the relay characteristics amplitude; use e and $\dot{e}$ as the phase plane coordinates, and also consider

$$e = r - c \quad (6-2)$$

$$r = R \cdot 1(t), \quad \dot{e} = -\dot{c} \quad (6-3)$$

Then (8-1) can be changed as

$$T\ddot{e} + \dot{e} + KM = 0 \quad e > 0$$

$$T\ddot{e} + \dot{e} - KM = 0 \quad e < 0 \quad (6-4)$$

The phase track curve of this system is as shown in Figure 6.3:

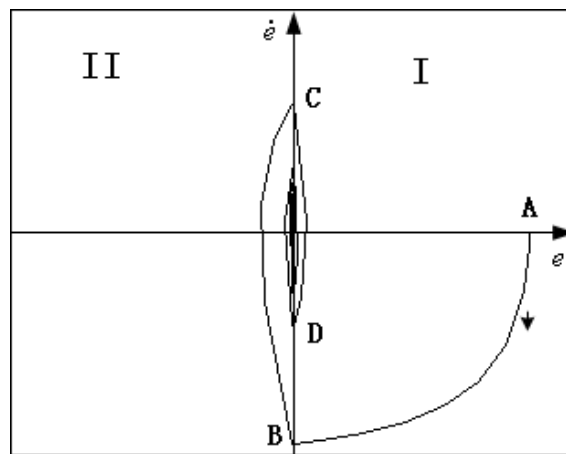


Figure 6.3

Observe that X_1 is $-e$, X_2 is $-\dot{e}$; take coordinates of X_1 - X_2 can get phase track $(-e, -\dot{e})$; when give coordinates exchange we get phase track $(e, \dot{e})$.

- **Relay-Type Non-Linear Closed-Loop System with Velocity Negative Feedback.**

The principle block diagram of relay-type non-linear closed-loop system velocity negative feedback is shown as in Figure 6.4.

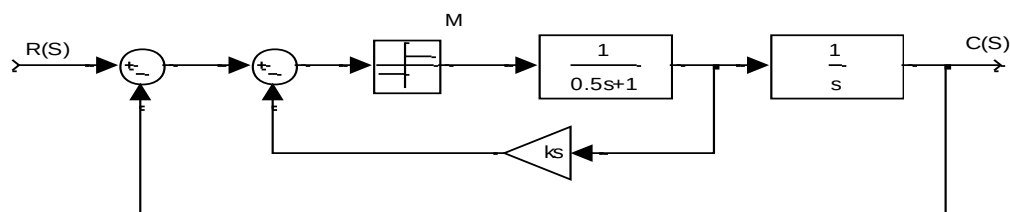


Figure 6.4

Its analog circuit is shown as in Figure 6.5.

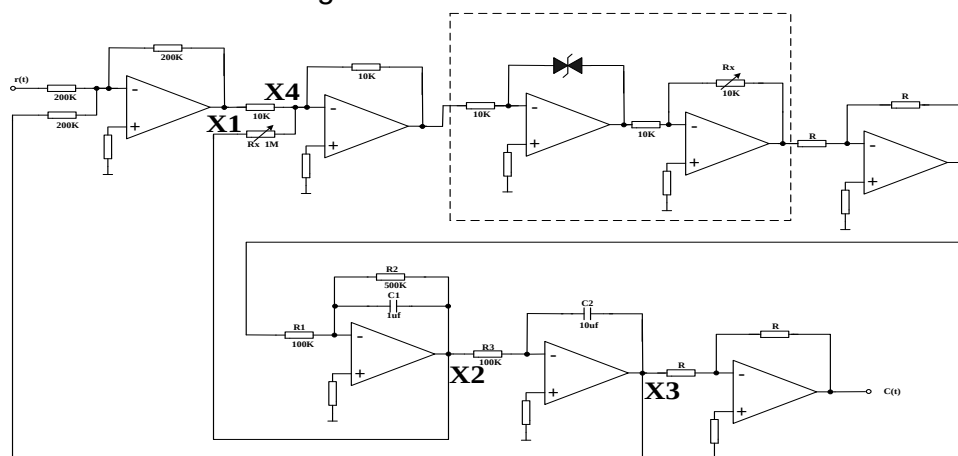


Figure 6.5

When construct this diagram, can add some inverted-phase parts when necessary; when

feedback part multiplies with output $c(t)$ can get $T\dot{c}$; then bring out from so from X_2 and observe that X_1 is $-e$, X_2 is $-\dot{e}$; take X_1 - X_2 coordinates can get phase track $(-e, -\dot{e})$; when give coordinates inverted phase change can get the coordinates of $(e, \dot{e})$. The phase track curve of this system is as shown in Figure 6.6:

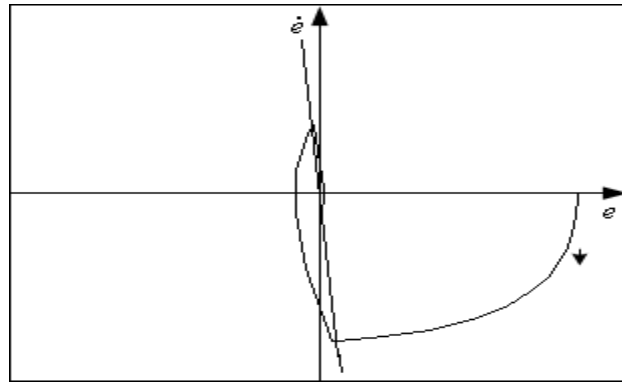


Figure 6.6

Graph line in this figure will appear and its boundary equation is: $e - k_s \dot{e} = 0$, where k_s as the feedback coefficient. When increase the feedback resistance, the phenomenon will be more apparent.

- **Saturation-type Non-Linear Closed-Loop System.**

The principle block diagram of saturation-type non-linear closed-loop system is shown as in Figure 8.7.

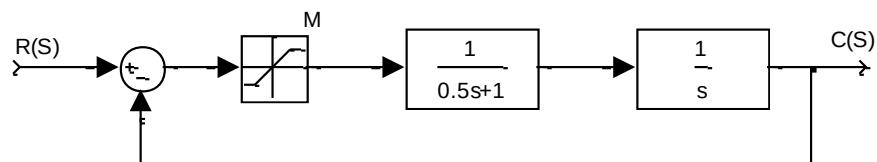


Figure 6.7

Its analog circuit is shown as in Figure 6.8.

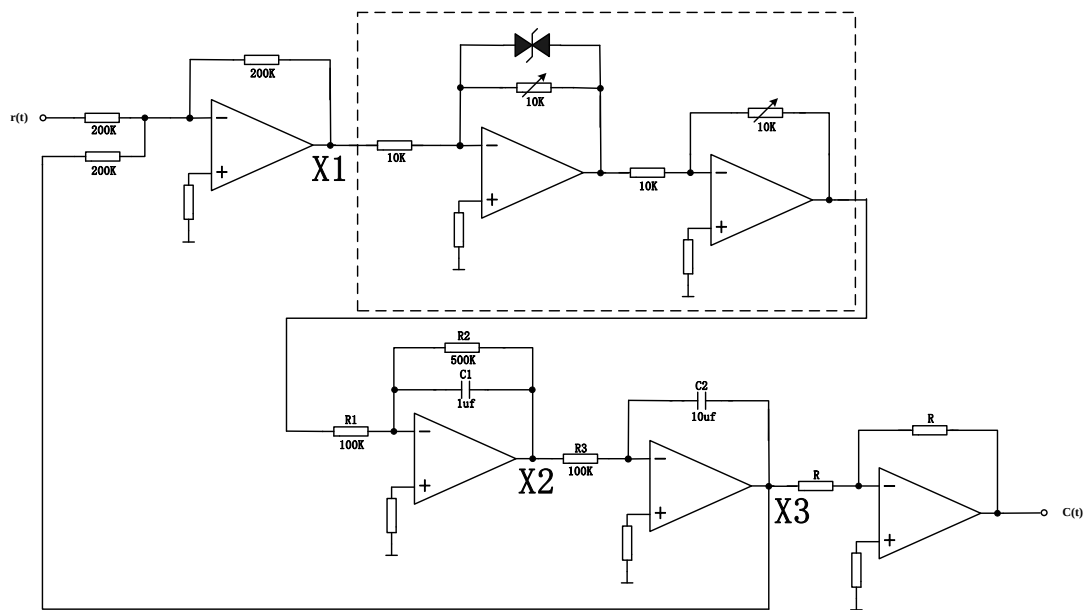


Figure 6.8

The circuit of system shown can be described as following equations:

$$T\ddot{e} + \dot{e} + e = 0 \quad |e| < M$$

$$T\ddot{e} + \dot{e} + M = 0 \quad e > M$$

$$T\ddot{e} + \dot{e} - M = 0 \quad e < -M \quad (6-5)$$

With the same principles can be observed that X_1 is $-e$, X_2 is $-\dot{e}$; take X_1 - X_2 coordinates can get phase track $(-e, -\dot{e})$; when give coordinates inverted phase change can get the coordinates $(e, \dot{e})$. The phase track curve of this system is as shown in Figure 6.9:

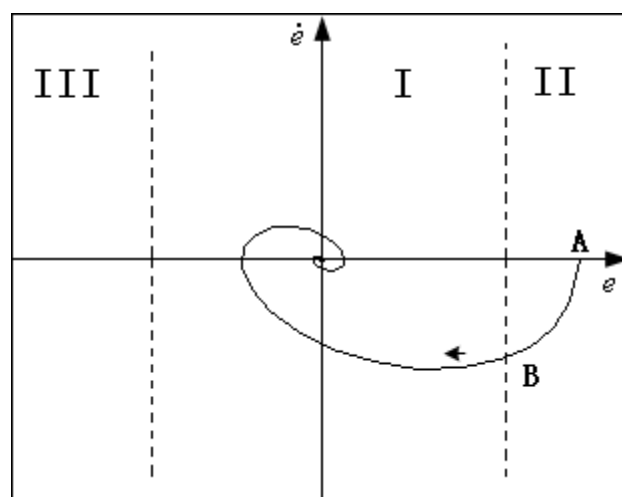


Figure 6.9

PROCEDURES:-

- According to experiment principles to design and connect an analog circuit of relay-type non-linear closed-loop system without calibration with experiment apparatuses; through input step response of different amplitudes to observe the phase track of the system on phase plane; draw track diagram and analyze step response and steady-state error of the system according to the these phase tracks. This experiment can use op-amp units I, II, III, VI, non-linear part I and capacitances of Component Unit.
- Press down Button Switches SW2 and SW3 of Power Unit at the same time, then press down SW5; adjust adjustable potentiometer P1 to make -12V-+12 V output voltage to be +1 V thus form unit step signal circuits, and then press down SW2 and SW3 again, then power off.
- Connect circuit according to Figure 6.2; press down the Button Switches used in op-amp unit and non-linear part.
- Set oscilloscope in X-Y Test file; connect the oscilloscope input side X with X1 side shown in Figure 6.2; connect Y input side with X2 side shown in Figure 6.2.
- When press down Button Switches SW2 and SW3 of Power Unit at the same time, observe input-output display of relay-type non-linear closed-loop system without calibration with oscilloscope and record down the results.
- Readjust P1 to change the input step signals and repeat procedures (3) and (4), and record down the results.
- According to experiment principles to design and connect an analog circuit of relay-type non-linear closed-loop system with velocity negative feedback with experiment apparatuses; through input step response of different amplitudes to observe the phase track of the system on phase plane; draw track diagram and analyze step response and steady-state error of the system according to the these phase tracks, and also compare with relay-type non-linear closed-loop system without calibration. This experiment can use op-amp units I, II, III, IV, V, VI, non-linear part I and capacitances of Component Unit.
- Press down Button Switches SW2 and SW3 of Power Unit at the same time, then press down SW5; adjust adjustable potentiometer P1 to make -12V-+12 V output voltage to be +1 V thus form unit step signal circuits, and then press down SW2 and SW3 again, then power off.
- Connect circuit according to Figure 6.5; press down the Button Switches used in op-amp unit and non-linear part.
- Set oscilloscope in X-Y Test file; connect the oscilloscope input side X with X1 side shown in Figure 6.5; connect Y input side with X2 side shown in Figure 6.5.
- When press down Button Switches SW2 and SW3 of Power Unit at the same time, observe input-output display of relay-type non-linear closed-loop system with velocity negative feedback with oscilloscope and record down the results.
- Readjust P1 to change the input step signals and repeat procedures (3) and (4), and record down the results.
- According to experiment principles to design and connect an analog circuit of saturation-type non-linear closed-loop system with experiment apparatuses; through input step response of different amplitudes to observe the phase track of the system on

phase plane; draw track diagram and analyze step response and steady-state error of the system according to the these phase tracks. This experiment can use op-amp units I, II, III, VI, non-linear part II and capacitances of Component Unit.

- Press down Button Switches SW2 and SW3 of Power Unit at the same time, then press down SW5; adjust adjustable potentiometer P1 to make -12V-+12 V output voltage to be +1 V thus form unit step signal circuits, and then press down SW2 and SW3 again, then power off.
- Connect circuit according to Figure 6.8; press down the Button Switches used in op-amp unit and non-linear part.
- Set oscilloscope in X-Y Test file; connect the oscilloscope input side X with X1 side shown in Figure 8.8; connect Y input side with X2 side shown in Figure 6.8.
- When press down Button Switches SW2 and SW3 of Power Unit at the same time, observe input-output display of saturation-type non-linear closed-loop system with oscilloscope and record down the results.
- Readjust P1 to change the input step signals and repeat procedures (3) and (4), and record down the results.
- Analyze experiment results and complete experiment report.

RESULTS:-

- Draw analog circuits of relay-type, relay-type with velocity negative feedback and saturation-type non-linear system.
- Draw the phase tracks of relay-type, relay-type with velocity negative feedback and saturation-type non-linear system in graphical method.
- Observe the phase tracks on phase plane of relay-type, relay-type with velocity negative feedback and saturation-type non-linear system, then compare the observed phase tracks with drawn ones. Analyze the differences and steady-state error between the two.

THINKING QUESTIONS:-

- How to get c and $\dot{c}$ signals for the experiments? And how to get e and $\dot{e}$ signals?
- Try to explain the relationships between phase track($e, \dot{e}$) and phase track($c, \dot{c}$).

EXPERIMENT

NO.7

EXPERIMENT NO:7

NAME:-

Pole Configuration Whole State Feedback Control Circuit Analog Experiment.

OBJECTIVE:-

- Study and master the method of designing whole state feedback control system with pole configuration method.
- Research on the effect of parameter to system capability with circuit analog method.
- Research on method of using state feedback configure pole in modern control theory.

APPARATUSES:-

- CL-PID: Op-Amp Application Trainer Kit.
- Low frequency slow-scan oscilloscope.
- Digital Multi-meter.
- Connecting Links.

THEORY:-

Experimental Contents

- Design typical two-order system pole configuration whole state control system, and do the experiment according to designed parameters, compare the results before and after feedback.
- Design typical three-order system pole configuration whole state control system, and do the experiment according to designed parameters, compare the results before and after feedback.

Experimental Principles

1. Typical Two-Order System Whole State Feedback Pole Configuration Design

Method.

If the controlled system totally controllable, then we can configure poles via state feedback, showed as following system block diagram:

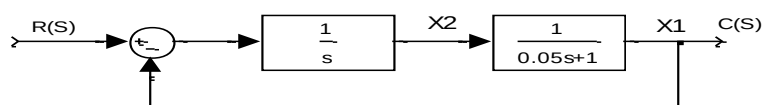
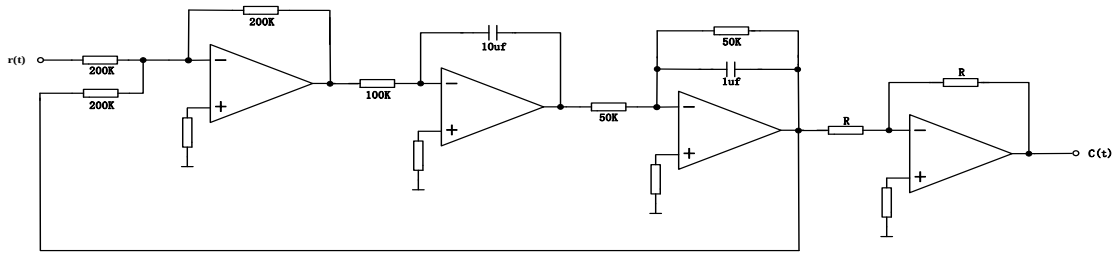


Figure 7.1

The system's analog circuit figure showed as Figure 7.2:

**Figure 7.2**

Imagining expected capabilities parameters: overshoot $M_p \leq 5\%$, peak time $t_p \leq 0.5$.

Count expected pole according to capabilities parameters: $p_1 = -7.07 + j7.07$,

$$p_2 = -7.07 - j7.07.$$

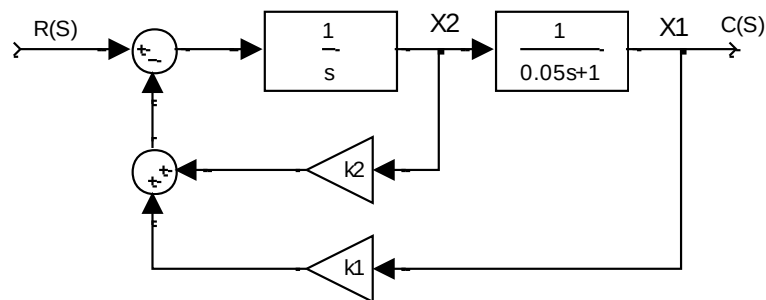
Original system open loop transformation function $G(S) = \frac{1}{S(0.05S + 1)}$, set X_1 , X_2 为 as

state variable in the block diagram, make the system open loop transformation function to be state equation, we can get:

$$\dot{X} = \begin{pmatrix} -20 & 20 \\ -1 & 0 \end{pmatrix} X + \begin{pmatrix} 0 \\ 1 \end{pmatrix} u$$

$$y = (1 \ 0)X$$

After designing of pole configuration, system block diagram showed as Figure 9.3, count out system state feedback matrix $K = [10.9 \ -5.9]$.

**Figure 7.3**

The system analog circuit diagram showed as Figure 7.4:

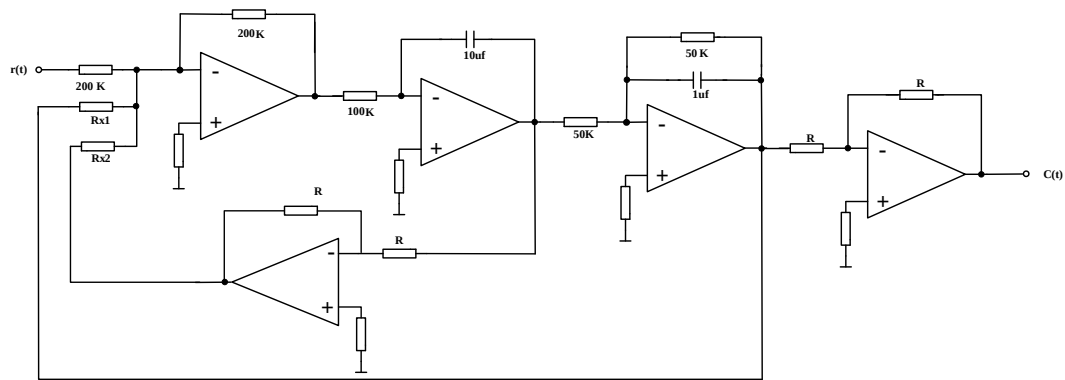


Figure 7.4

Parameter R_{x1} and R_{x2} is 18.3K and 33.9 K. (Pay attention to feedback circuit's connection) Step response curve diagrams before and after system configuration showed as following Figure 7.5 and Figure 7.6:

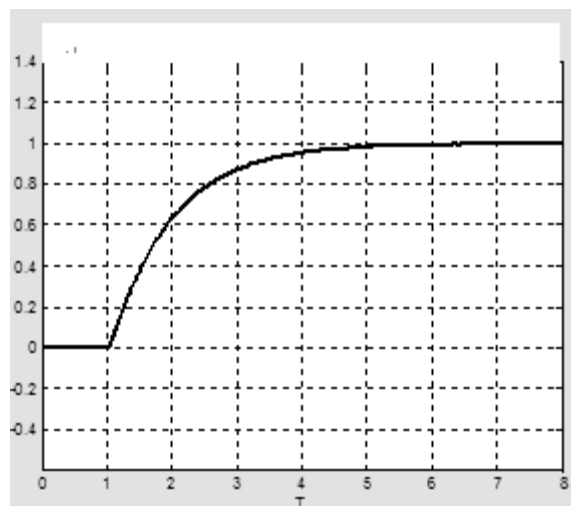


Figure 7.5

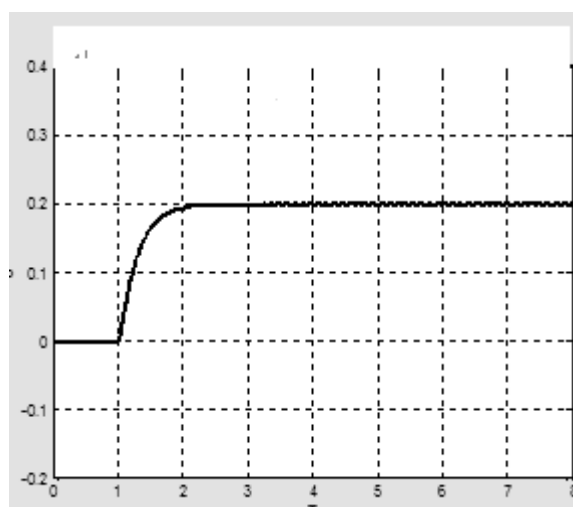


Figure 7.6

2. Typical third-order system whole state feedback pole configuration design method.

2.1 Typical third-order system block diagram showed as Figure 7.7:

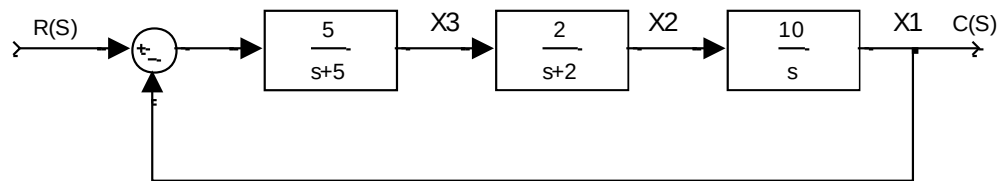


Figure 7.7

Its open loop transformation function is $G(S) = \frac{100}{S(S+5)(S+2)}$

Closed loop transformation function is $W(S) = \frac{G(S)}{1+G(S)}$

Analog circuit diagram showed as following Figure 7.8:

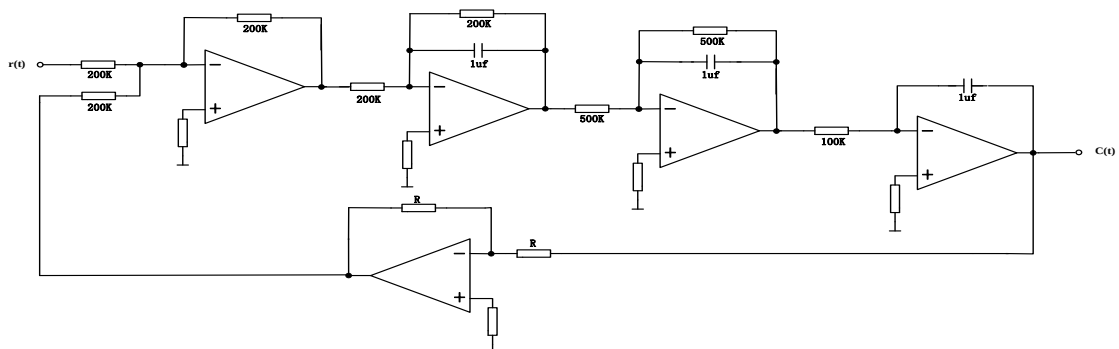


Figure 7.8

We can use Louts criterion to judge the system is not stable, its unit step response curve

Showed as following Figure7.9:

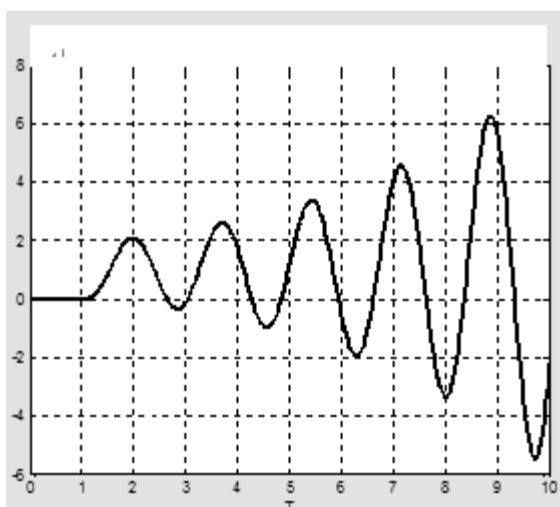


Figure 7.9

Choose the values in X_1 , X_2 , X_3 as state variable in the diagram, system open loop

transformation function can be expressed as: $A = \begin{pmatrix} 0 & 10 & 0 \\ 0 & -2 & 2 \\ 0 & 0 & -5 \end{pmatrix}$, $B = \begin{pmatrix} 0 \\ 0 \\ 5 \end{pmatrix}$

$$y = (1 \ 0 \ 0)X$$

2.2 Ideal pole and state equation

After considering about stability requests, we choose ideal pole as $S_1 = -9$, $S_2 = -2 + j$,

$S_3 = -2 - j$, so the ideal closed loop pole characteristic equation is

$$S^3 + 13S^2 + 44S + 72 = 0$$

2.3 Whole state feedback coefficient design:

Choose the values in X_1 , X_2 , X_3 as state variable in the diagram, the block diagram of typical third-order system with whole state feedback can be expressed as Figure 9.10:

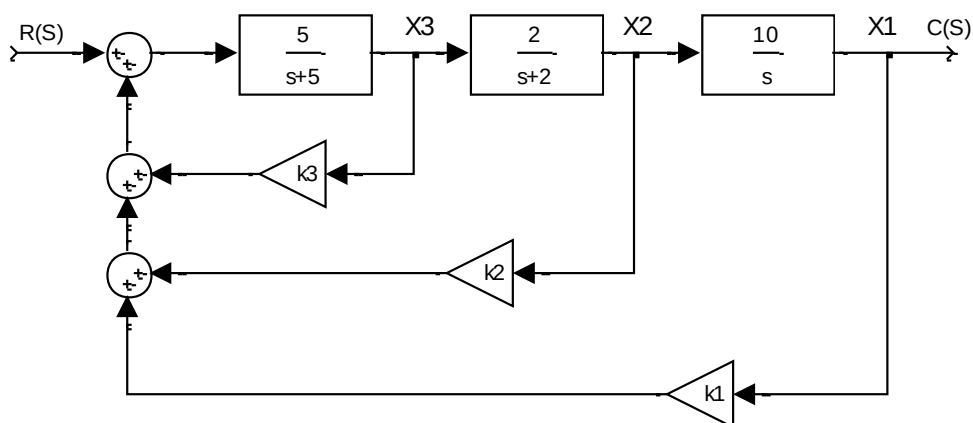


Figure 7.10

Solving plus whole state feedback closed loop characteristic equation, we can get:

$$S^3 + (7 - 5k_3)S^2 + (10 - 10k_2 - 10k_3)S - 100k_1 = 0$$

Let it to be correspondence with ideal closed loop characteristic equation, we can get whole state feedback coefficient $k_1 = -0.72$, $k_2 = -0.22$, $k_3 = -1.2$.

2.4 Typical system of whole state feedback

Analog circuit of typical system for whole state feedback showed as Figure 9.11.

Resistances of parameter R_{x1} , R_{x2} , R_{x3} are 278K, 91K, and 167K.

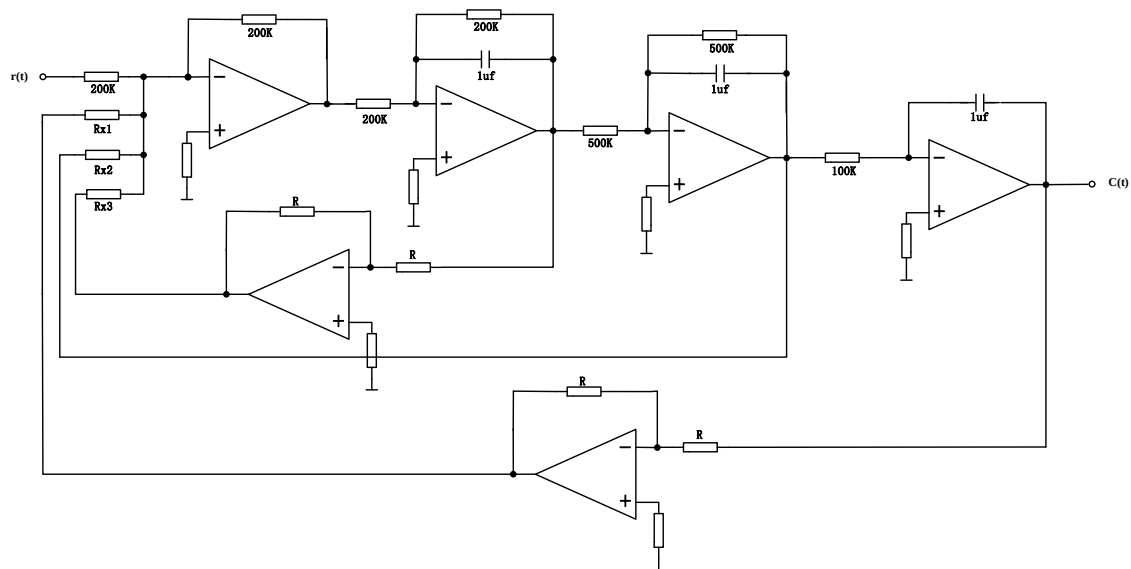


Figure 7.11

After configuration, unit step response curve of the system showed as Figure 7.12.

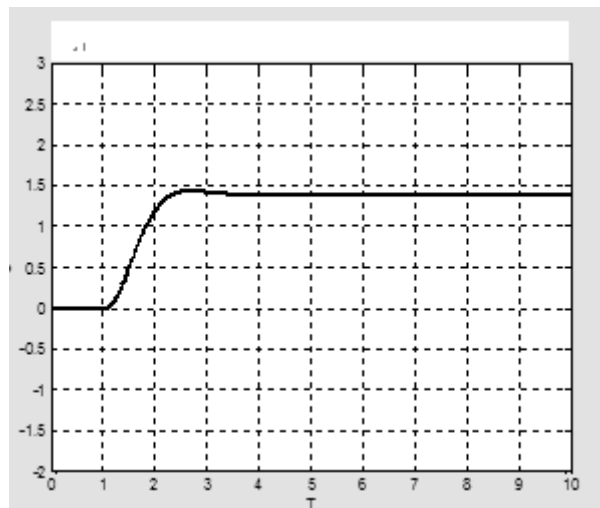


Figure 7.12

PROCEDURES:-

- **Typical Second Order System.**
- Design a typical second order system according to experiment principle with experimental apparatus. The second order system can use operational amplifier unit I, II, III, IV, etc.
- Press the keyswitches SW2 and SW3 at the same time, then press SW5, adjust adjustable potentiometer P1, make the output voltage of -12V—+12V to be +1V, forming unit step signal circuit, then Press the keyswitches SW2 and SW3 at the same time again to shut off the power supply.
- Press the keyswitches of operational amplifier unit which can be used in circuit.
- Use wire to connect input terminal of analog circuit and IN10, connect output terminal and oscilloscope.
- When we press the keyswitches SW2 and SW3 at the same time, observe output

terminal step response curve with oscilloscope, record the results waveforms.

- Use pole configuration method to design typical second order system's whole state feedback coefficients.
- Design according to designed parameters and connect the analog circuit according to Figure 9.4, the second order system can use operational amplifier unit I, II, III, IV and other component unit capacitance, adjustable resistance. Repeat steps 1.2, 1.3, 1.4, 1.5 to observe its step response, then record the result waveforms.
- Change feedback coefficient, repeat steps 1.2, 1.3, 1.4, 1.5 to observe the effect of its step response.
- Analyze experiment results, finish experiment report.

- **Typical Three Order System.**

- Design a typical third order system according to experiment principle with experimental apparatus. The third order system can use operational amplifier unit I, II, III, V, VI and other component unit.
- Press the keyswitches SW2 and SW3 at the same time, then press SW5, adjust adjustable potentiometer P1, make the output voltage of $-12V$ — $+12V$ to be $+1V$, forming unit step signal circuit, then Press the keyswitches SW2 and SW3 at the same time again to shut off the power supply.
- Connect circuit according to Figure 9.8, press the keyswitches of operational amplifier unit which can be used in circuit.
- Use wire to connect input terminal of analog circuit and IN10, connect output terminal and oscilloscope.
- When we press the keyswitches SW2 and SW3 at the same time, observe output terminal step response curve with oscilloscope, record the results waveforms.
- Use pole configuration method to design typical third order system's whole state feedback coefficients.
- Design according to designed parameters and connect the analog circuit according to Figure 9.11, the third order system can use operational amplifier unit I, II, III, V, VI, VII and other component unit. Repeat steps 2.2, 2.3, 2.4, 2.5 to observe its step response, then record the result waveforms.
- Change feedback coefficient, repeat steps 2.2, 2.3, 2.4, 2.5 to observe the effect of its step response.
- Analyze experiment results, finish experiment report.

RESULTS:-

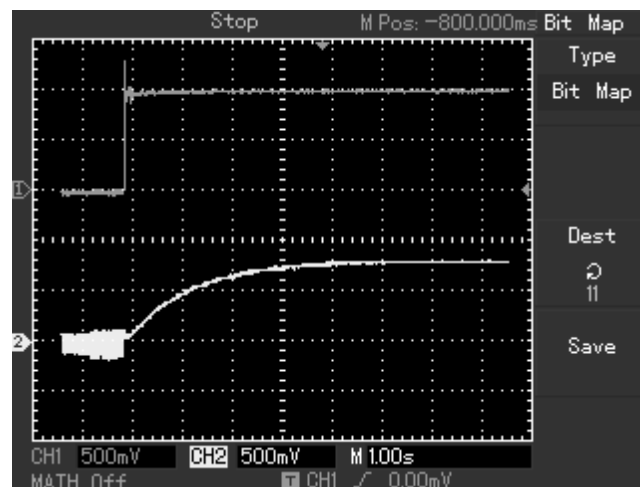
- Draw second order, third order system analog circuit diagram according to designing, and draw the step response curve diagram of the system being tested.
- Use pole configuration method to design and count out the whole state feedback coefficients of each system.
- Design and draw the feedback analog circuit of each system according to counted feedback coefficient then draw each feedback system's step response curve diagram.
- Analyze and compare step response diagram before and after feedback, study the effect of feedback parameter to each capability of the system.

THINKING QUESTIONS:-

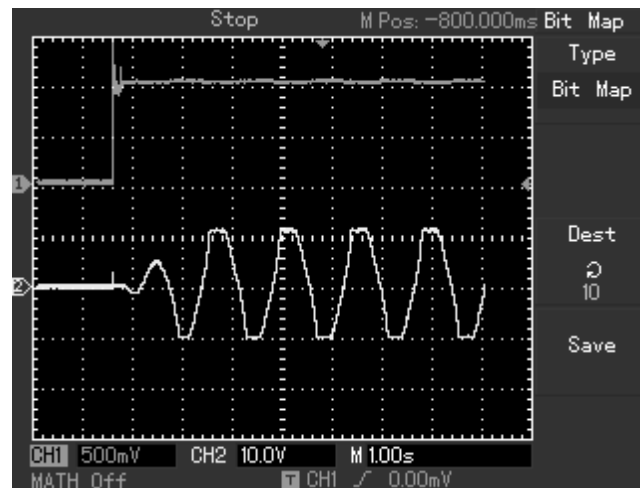
- How to confirm whether a system controllable?
- Whether it has affected on system zero point during state feedback changing system pole?
- Try to analyze the effect of feedback structure to system capability.

CAPTURED WAVEFORM:-

- 1) 2nd order system whole state feedback pole configuration design.



2) 3rd order system whole state feedback pole configuration design.



3) Whole state feedback coefficient design:

